Layer-resolved G_0W self-energy of van der Waals heterostructures and the quantum-twisting-microscope conductance

Theory and implementation notes for the `g0w_heterostructure` package

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Abstract

These notes document, in a self-contained way, the physics and the numerics implemented in the `g0w` package: the layer-resolved Coulomb kernel of N two-dimensional sheets in a stratified, possibly gated and dispersive dielectric; the random-phase-approximation (RPA) dielectric matrix and screened interaction of a stack of decoupled electronic subsystems; the one-shot G_0W (correlation) self-energy in the full-frequency and plasmon-pole forms, including its band-matrix generalisation for multiband samples; the Dyson equation and the electronic spectral function; and the differential conductance of a quantum twisting microscope (QTM) in the limit of a monolayer-graphene tip with a small Fermi surface. Two samples are worked out in detail: doped monolayer graphene (SLG | hBN | SLG) and twisted bilayer graphene (SLG tip | hBN | TBG). Every equation is followed by the name of the function that implements it, and every approximation is listed explicitly.

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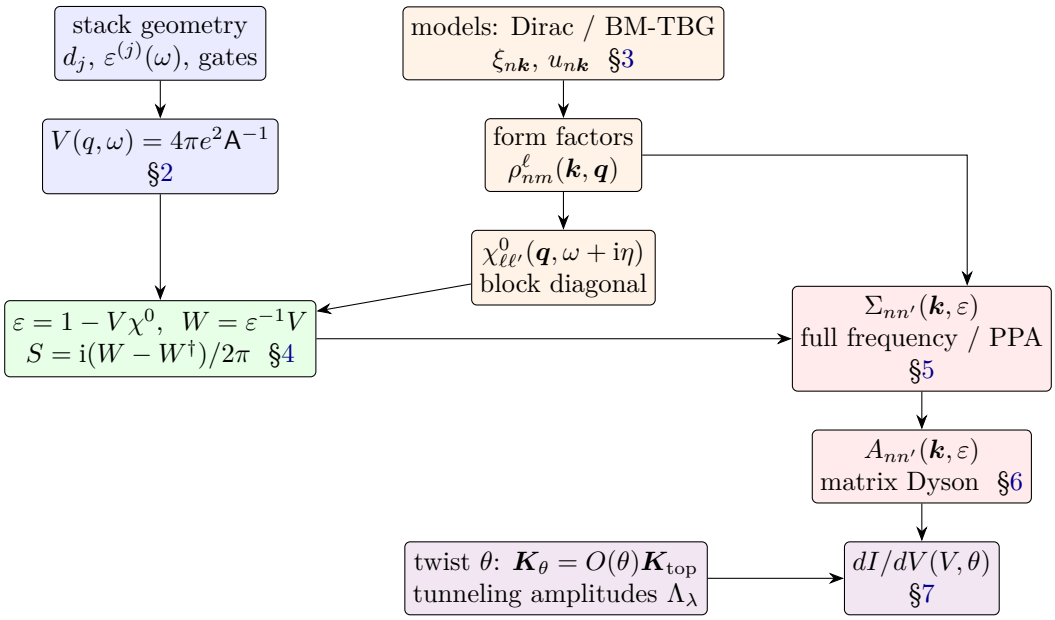


Figure 1: Computational chain. Blue: electrostatics of the stack. Orange: electronic structure of the subsystems. Green: RPA screening. Red: G_0W self-energy and spectral function. Violet: QTM.

1 Overview

1.1 What is computed

The package computes the single-particle spectral function of the electrons of a two-dimensional *sample* embedded in an arbitrary van der Waals stack, dressed by the dynamically screened Coulomb interaction at the level of one-shot G_0W, and the momentum-resolved tunneling conductance that a quantum twisting microscope (QTM) measures on it. The chain of quantities is shown in fig. 1:

1. the **bare Coulomb kernel** $V_{\ell\ell'}(q,\omega)$ between the N sheets of the stack, from Poisson's equation in a stratified uniaxial dielectric with optional gates (section 2),
2. the **electronic model** of every subsystem: bands $\xi_{n\boldsymbol{k}}$, Bloch states and the layer-projected density matrix elements (form factors) $\rho^{\ell}_{nm}(\boldsymbol{k},\boldsymbol{q})$ (sections 3 and 5.2),
3. the **independent-particle polarisability** $\chi^0_{\ell\ell'}(\boldsymbol{q},\omega)$, the RPA dielectric matrix $\varepsilon = 1 - V\chi^0$ and the **screened interaction** $W = \varepsilon^{-1}V$, with its spectral density S, the loss function and the collective modes (section 4),
4. the **retarded G_0W self-energy** $\Sigma_{nn'}(\boldsymbol{k},\varepsilon)$ of the sample (section 5),
5. the **spectral function** $A_{nn'}(\boldsymbol{k},\varepsilon)$ from the Dyson equation (section 6),
6. the **QTM differential conductance** $dI/dV(V,\theta)$ and d^2I/dV^2 (section 7).

The results of the worked examples (notebooks 01–05) are collected in section 8 and interpreted in section 9 (notebook 06). Grids, cost, validation and limitations are discussed in section 10.

1.2 Units and conventions

quantity	convention
energies, frequencies, temperature	eV, with $\hbar = k_B = 1$
lengths, momenta	nm, nm^{-1}
electrostatics	Gaussian: $\nabla \cdot (\boldsymbol{\varepsilon} \nabla \phi) = -4\pi\rho$, $e^2 = 1.4399764\ \mathrm{eV\,nm}$
2D Fourier transforms	$f(\boldsymbol{q}) = \int \mathrm{d}^2 r\, e^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} f(\boldsymbol{r})$, so that $V(q)$ is in $\mathrm{eV\,nm}^2$
momentum integrals	$\frac{1}{A}\sum_{\boldsymbol{q}} \to \int \frac{\mathrm{d}^2 q}{(2\pi)^2} \simeq \sum_{\boldsymbol{q}} w_{\boldsymbol{q}}$, A the sample area, quadrature weights $w_{\boldsymbol{q}}$ in nm^{-2} (section 5.9)
occupations	$n_{\mathrm{F}}(x) = 1/(e^{\beta x}+1)$, $n_{\mathrm{B}}(x) = 1/(e^{\beta x}-1)$, $\beta = 1/k_B T$
response	retarded, evaluated at $\omega + \mathrm{i}\eta$, $\eta > 0$
dielectric matrix	$\varepsilon = 1 - V\chi^0$ $\quad$ (*not* $1 - \chi^0 V$)
screened interaction	$W = \varepsilon^{-1} V$ $\quad$ (*not* $V\varepsilon^{-1}$)
spectral density of W	$S = \mathrm{i}(W - W^\dagger)/(2\pi)$, $= -\operatorname{Im} W/\pi$ for symmetric W
energies of states	$\xi_{n\boldsymbol{k}} = E_{n\boldsymbol{k}} - \mu$ (measured from the sample Fermi level)
spectral function	$A = \mathrm{i}(G - G^\dagger)/(2\pi)$, $\int A\,\mathrm{d}\varepsilon = 1$ per state
twist rotation	$O(\theta) = \left(\begin{smallmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{smallmatrix}\right)$ (clockwise, as in Ref. [1])
flattened index	$(\ell, \boldsymbol{G}) \mapsto \ell\, n_G + G$ for every matrix in (layer, reciprocal-vector) space

1.3 Physical assumptions

The following assumptions are made throughout. Where one of them is relaxed or tested, this is said explicitly.

Assumption 1 (Electrostatics)**.** *The sheets are infinitely thin and the Coulomb interaction is quasi-static (no retardation: $q \gg \omega/c$). Each dielectric region is homogeneous, local and uniaxial with the optical axis along z, $\boldsymbol{\varepsilon} = \mathrm{diag}(\varepsilon_\parallel, \varepsilon_\parallel, \varepsilon_z)$, possibly dispersive and lossy. Gates are grounded perfect conductors.*

Assumption 2 (Decoupled subsystems)**.** *Different electronic subsystems (e.g. the QTM tip and the sample) are not tunnel coupled when the screening is computed: there is no density operator connecting them, so χ^0 is block diagonal and all inter-subsystem physics is in V. Within a subsystem (the two layers of TBG) the layers are hybridised by the electronic Hamiltonian and χ^0 has layer-off-diagonal elements.*

Assumption 3 (RPA and one-shot $\mathrm{G_0W}$)**.** *The density response is the RPA one built on the bare (or Hartree) bands, and the self-energy is the first term of Hedin's equations, $\Sigma = \mathrm{i}G_0 W_0$, with the non-interacting G_0 of the sample. Vertex corrections and self-consistency are neglected. Only the* correlation *part $\Sigma_c = \mathrm{i}G_0(W - V)$ is kept, and the bare exchange (Fock) term is excluded (section 5.12).*

Assumption 4 (Head only)**.** *Unless stated otherwise only the $\boldsymbol{G} = \boldsymbol{G}' = 0$ (head) element of the moiré reciprocal-lattice structure of χ^0, W and of the form factors is kept (local-field effects off, $n_G = 1$). In the self-energy this also restricts the momentum transferred to the electron to the first moiré Brillouin zone.*

Assumption 5 (Weak tunneling and small tip Fermi surface)**.** *Tip–sample tunneling is treated to lowest order and conserves in-plane momentum. The tip is a non-interacting graphene sheet whose chemical potential follows the bias while its bands stay fixed relative to the sample Fermi level, and the sample chemical potential does not move relative to the sample bands. The tip Fermi momentum is small compared with the momentum scale on which the sample spectral function varies. The temperature is zero, and finite temperature is mimicked by smoothing. Only elastic tunneling into the dressed sample states is included (interactions enter through A^S) [1].*

1.4 Package layout

module	content
`g0w.layer_coulomb`	$V(q,\omega)$ of N sheets, gates, capped slab, branch-tracked $\kappa(\omega)$ (§2)
`g0w.dielectrics`	hBN, SiO_2, Al_2O_3 dielectric functions
`scripts/plot_notes_selfenergy.py`	figures of section 5 (quadrature grids, anatomy of Σ)
`g0w.layer_helpers`	`layer_helpers_G`, `LayerHelpers`: ρ^{ℓ}_{nm}, χ^0, ε, loss from `pybandstructure`
`g0w.screening`	block-diagonal χ^0, ε and W of a stack, loss, layer-channel split
`g0w.self_energy_ppa`	plasmon poles and residues, PPA self-energy, q weights
`g0w.self_energy_full_frequency`	full-frequency Σ, band-matrix $\Sigma_{nn'}$, resolvent kernels
`g0w.spectral`	scalar/matrix Dyson equation, A, quasiparticle solution
`g0w.qtm`	twist kinematics, tunneling amplitudes, dI/dV, d^2I/dV^2 (SLG and TBG)
`g0w.models.dirac_chi`	analytic χ^0 of doped graphene
`g0w.models.slg_stack`, `slg_self_energy`	SLG|hBN|SLG junction, Dirac form factor, Σ, A
`g0w.models.tbg_model`, `tbg_stack`, `tbg_self_energy`	TBG continuum model, three-sheet stack, $\Sigma_{nn'}$
`g0w.satellites`	satellite energies: emission edge, energy-loss decomposition of $-\operatorname{Im}\Sigma$ (§9)
`notebooks/01–06`	executed examples: geometry, SLG G_0W, TBG G_0W, QTM on SLG and on TBG, satellites
`scripts/run_slg_qtm.py`, `run_tbg_qtm.py`	command-line drivers for the dI/dV maps (`.npz` + `.png`)
`tests/`	regression and physics tests (`python tests/run_tests.py`)

2 Electrostatics of the stack: the layer-resolved Coulomb kernel

The kernel $V_{\ell m}(q,\omega)$ defined in this section is the interaction energy of two electrons placed on sheets ℓ and m, as a function of their in-plane wavevector. It is the quantity that enters the interaction Hamiltonian (23). It is obtained by solving Poisson's equation in the stratified medium, in the spirit of the quasi-static treatments of double layers and of van der Waals heterostructures [25, 24].

2.1 Geometry

N sheets $\ell = 0, \dots, N-1$ sit at heights $z_0 > z_1 > \cdots > z_{N-1}$ and divide space into $N+1$ regions (fig. 2): region 0 above sheet 0, region j ($1 \le j \le N-1$) between sheets $j-1$ and j, of thickness $d_{j-1} = z_{j-1} - z_j$ (`d[j-1]` in the code), and region N below sheet $N-1$. The outer regions are half-spaces, or are terminated by grounded gates at distances d_{t} above sheet 0 and d_{b} below sheet $N-1$ (`top_gate_d`, `bottom_gate_d`), or (top only) by a slab of thickness d_{s} backed by a half-space ε_{c} (`CappedLayerCoulomb`).

2.2 Poisson equation in one region

With $\boldsymbol{E} = -\nabla\phi$, $\boldsymbol{D} = \boldsymbol{\varepsilon}\boldsymbol{E}$ and $\nabla\cdot\boldsymbol{D} = 4\pi\rho$, a 2D Fourier transform in the plane gives, inside region j (no free charge),

$$\varepsilon_z^{(j)}\partial_z^2\phi - \varepsilon_{\parallel}^{(j)} q^2\phi = 0 \quad \Rightarrow \quad \phi'' = \kappa_j^2\phi, \qquad \kappa_j = q\sqrt{\varepsilon_{\parallel}^{(j)}/\varepsilon_z^{(j)}}, \qquad Y_j = \varepsilon_z^{(j)}\kappa_j = q\sqrt{\varepsilon_{\parallel}^{(j)}\varepsilon_z^{(j)}}. \quad (1)$$

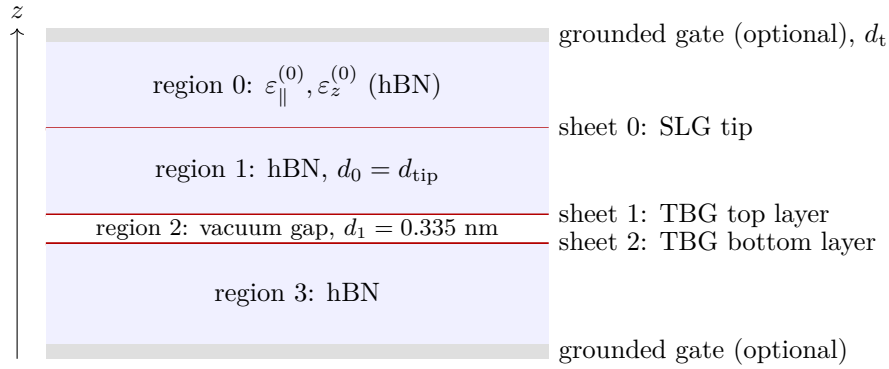


Figure 2: The three-sheet stack used for the TBG sample (notebooks 03, 05). For SLG|hBN|SLG there are two sheets and three hBN regions.

Y_j is the *admittance* of the region (`compute_kappa_Y`). For a spacer of thickness d whose lower and upper faces are at potentials Φ_b and Φ_a, the solution $\phi(z) = [\Phi_a \sinh \kappa z + \Phi_b \sinh \kappa(d-z)]/\sinh \kappa d$ $(0 \le z \le d)$ gives the normal displacement on the two faces:

$$\varepsilon_z \phi'(d^-) = Y\big[\Phi_a \coth \kappa d - \Phi_b \operatorname{csch} \kappa d\big], \qquad \varepsilon_z \phi'(0^+) = Y\big[\Phi_a \operatorname{csch} \kappa d - \Phi_b \coth \kappa d\big]. \tag{2}$$

For a half-space above sheet 0, $\phi = \Phi_0 e^{-\kappa(z-z_0)}$ and $\varepsilon_z \phi' = -Y_0 \Phi_0$. With a grounded gate at distance d_t, $\phi = \Phi_0 \sinh \kappa(z_g - z)/\sinh \kappa d_t$ and $\varepsilon_z \phi' = -Y_0 \coth(\kappa d_t)\, \Phi_0$. A gate therefore replaces the boundary admittance $Y \to Y \coth(\kappa d_g)$, which tends to Y for $d_g \to \infty$. A slab (κ_s, Y_s, d_s) capped by a half-space with $Y_c = \varepsilon_c q$ gives instead

$$Y_{\text{eff}} = Y_s \, \frac{Y_c + Y_s \tanh \kappa_s d_s}{Y_s + Y_c \tanh \kappa_s d_s}, \tag{3}$$

which interpolates between Y_c $(d_s \to 0)$, Y_s $(d_s \to \infty)$ and the gate result $(Y_c \to \infty)$ (`capped_slab_admittance`).

2.3 Matching at the sheets: the tridiagonal admittance matrix

The potential is continuous across sheet ℓ and the displacement jumps by the sheet charge, $D_z(z_\ell^+) - D_z(z_\ell^-) = 4\pi\sigma_\ell$, i.e. $-\varepsilon_z \phi'(z_\ell^+) + \varepsilon_z \phi'(z_\ell^-) = 4\pi\sigma_\ell$. Inserting eq. (2) for the regions above (region ℓ) and below (region $\ell+1$) sheet ℓ:

$$\big[Y_\ell \coth \kappa_\ell d_{\ell-1} + Y_{\ell+1} \coth \kappa_{\ell+1} d_\ell\big]\Phi_\ell - Y_\ell \operatorname{csch}(\kappa_\ell d_{\ell-1})\, \Phi_{\ell-1} - Y_{\ell+1} \operatorname{csch}(\kappa_{\ell+1} d_\ell)\, \Phi_{\ell+1} = 4\pi\sigma_\ell, \tag{4}$$

with the outermost terms replaced by the boundary admittances Y_{top} and Y_{bot}: Y_{top} is Y_0, $Y_0 \coth \kappa_0 d_t$ or Y_{eff}, and Y_{bot} is Y_N or $Y_N \coth \kappa_N d_b$. Here Φ_ℓ and σ_ℓ are the Fourier components of the potential and of the charge density on sheet ℓ. The system is $\mathsf{A}(q,\omega)\Phi = 4\pi\sigma$ with a tridiagonal admittance matrix A (called `A` in `build_tridiagonal_A_diagonals`):

$$\begin{aligned}
\mathsf{A}_{00} &= Y_{\text{top}} + Y_1 \coth \kappa_1 d_0, & \mathsf{A}_{N-1,N-1} &= Y_{N-1} \coth \kappa_{N-1} d_{N-2} + Y_{\text{bot}}, \\
\mathsf{A}_{\ell\ell} &= Y_\ell \coth \kappa_\ell d_{\ell-1} + Y_{\ell+1} \coth \kappa_{\ell+1} d_\ell, & & \\
\mathsf{A}_{\ell,\ell-1} &= -Y_\ell \operatorname{csch} \kappa_\ell d_{\ell-1}, & \mathsf{A}_{\ell,\ell+1} &= -Y_{\ell+1} \operatorname{csch} \kappa_{\ell+1} d_\ell.
\end{aligned} \tag{5}$$

A unit point charge e on sheet m has $\sigma_\ell(\boldsymbol{q}) = e\,\delta_{\ell m}$, and the energy of a second charge e on sheet ℓ is $e\Phi_\ell$. The layer-resolved Coulomb kernel is therefore

$$\boxed{V_{\ell m}(q,\omega) = 4\pi e^2 \,[\mathsf{A}^{-1}(q,\omega)]_{\ell m}} \qquad [\text{eV nm}^2] \tag{6}$$

(`LayerCoulomb.coulomb_matrix`). A is inverted with a Thomas LU factorisation and coth, csch are evaluated in the overflow-safe form $\coth z = (1+u)/(1-u)$, $\operatorname{csch} z = 2e^{-z}/(1-u)$, $u = e^{-2z}$, with $1 - u = -\text{expm1}(-2z)$.

Closed forms (tested). A single sheet in a uniform medium has $\mathsf{A} = 2\varepsilon q$, i.e. $V = 2\pi e^2/(\varepsilon q)$. With gates at $\pm d$ it becomes $V = \frac{2\pi e^2}{\varepsilon q}\tanh qd$, and with one gate at distance d, $V = \frac{2\pi e^2}{\varepsilon q}(1-e^{-2qd})$. Two sheets in a uniform uniaxial medium:

$$V_{11} = V_{22} = \frac{2\pi e^2}{\bar{\varepsilon} q}, \qquad V_{12} = V_{11}\, e^{-\kappa d}, \qquad \bar{\varepsilon} = \sqrt{\varepsilon_\parallel \varepsilon_z}, \quad \kappa = q\sqrt{\varepsilon_\parallel/\varepsilon_z}. \tag{7}$$

2.4 Bilayer channels and the effect of gates

For two equivalent sheets, V is diagonalised by $u_\pm = (1, \pm 1)/\sqrt{2}$ with eigenvalues $V_\pm = V_{11} \pm V_{12}$. At $qd \ll 1$, $V_+ \simeq 4\pi e^2/(\bar{\varepsilon} q)$ (the total charge sees the long-range Coulomb field) while $V_- \simeq V_{11}\kappa d = 2\pi e^2 d/\varepsilon_z$ is finite (a dipole layer). V_+ drives the in-phase (optical, $\omega \propto \sqrt{q}$) plasmon and V_- the out-of-phase (acoustic, $\omega \propto q$) plasmon. Gates at distance d_g cut the long-range part, $V_+ \to 4\pi e^2 d_\mathrm{g}/\varepsilon_z$ for $qd_\mathrm{g} \ll 1$ (gates at $\pm d_\mathrm{g}$), but leave V_- almost unchanged when $d \ll d_\mathrm{g}$: gating raises the *relative* weight of the acoustic channel by suppressing the optical one (notebook 01, test `test_gate_suppresses_in_phase_more_than_out_of_phase`).

2.5 Dispersive dielectrics and passivity

hBN is modelled with one Lorentz oscillator per axis, the standard description of its two optical-phonon bands [26, 27],

$$\varepsilon_\alpha(\omega) = \varepsilon_{\infty,\alpha} + \frac{\varepsilon_{0,\alpha} - \varepsilon_{\infty,\alpha}}{1 - \omega^2/\omega_{T,\alpha}^2 - \mathrm{i}\gamma_\alpha\omega/\omega_{T,\alpha}^2}, \tag{8}$$

where ε_0 and ε_∞ are the static and high-frequency constants, ω_T the transverse-optical phonon energy and γ its damping, with $(\varepsilon_0, \varepsilon_\infty, \omega_T, \gamma) = (6.41, 4.53, 168, 0.82)$ (in-plane) and $(3.0, 2.5, 94.2, 0.23)$ (out-of-plane), energies in meV (the values of `dielectrics.HBNModel`). Between ω_T and $\omega_L = \omega_T\sqrt{\varepsilon_0/\varepsilon_\infty}$ (the Reststrahlen bands, 94–103 meV and 168–200 meV) $\varepsilon_\parallel\varepsilon_z < 0$ and κ is almost imaginary: the substrate carries hyperbolic phonon polaritons and V acquires a large dissipative part. For the retarded kernel of a passive medium the anti-Hermitian part must be negative semidefinite, $-(V - V^\dagger)/2\mathrm{i} \succeq 0$. The branch of $\kappa_j(\omega)$ is the decaying one ($\mathrm{Re}\,\kappa > 0$) at the first frequency of a sweep and is then continued without jumps (`compute_kappa_Y_tracked`). In the examples of notebook 01 the continued and the principal branches coincide and both are passive. Static values ($\omega \to 0$) are used unless the phonon range is of interest (all calculations in notebooks 02–05).

3 Electronic models

3.1 Doped monolayer graphene

Near one Dirac point, $h(\boldsymbol{p}) = \hbar v_F(\tau p_x\sigma_x + p_y\sigma_y)$ with $\boldsymbol{p}$ measured from the Dirac point, $\hbar v_F = 0.658$ eV nm ($v_F = 10^6$ m/s) and $g = g_s g_v = 4$. The bands and spinors are

$$\xi_{\boldsymbol{p}}^s = s\hbar v_F|\boldsymbol{p}| - E_F, \qquad u_{\boldsymbol{p}}^s = \tfrac{1}{\sqrt{2}}(1,\, s\, e^{\mathrm{i}\phi_{\boldsymbol{p}}}), \qquad s = \pm 1, \tag{9}$$

where $\tau = \pm 1$ labels the valley, $\phi_{\boldsymbol{p}}$ is the polar angle of $\boldsymbol{p}$, $k_F = \sqrt{4\pi n/g}$ the Fermi momentum at electron density n and $E_F = \hbar v_F k_F$ the Fermi energy ($n = 10^{12}$ cm$^{-2} \Rightarrow k_F = 0.1772$ nm^{-1}, $E_F = 116.6$ meV). The overlap that dresses every Coulomb vertex is the chirality factor

$$F^{s's}(\boldsymbol{p}, \boldsymbol{p}') = |\langle u_{\boldsymbol{p}}^{s'}|u_{\boldsymbol{p}'}^{s}\rangle|^2 = \tfrac{1}{2}\big[1 + ss'\cos\angle(\boldsymbol{p}, \boldsymbol{p}')\big], \qquad \textstyle\sum_s F^{s's} = 1, \tag{10}$$

which forbids intraband backscattering and interband forward scattering (`slg_self_energy.make_dirac_form_factor`) [28, 19]. The sample has only one sheet, so the layer-projected form factor of section 5.2 is $\rho^\ell = \delta_{\ell S}\langle u|u'\rangle$, where S labels the sample sheet: a sample electron has no intermediate state in the tip.

3.2 Twisted bilayer graphene: continuum model

Hamiltonian. One valley ($\tau = +$) of the Bistritzer–MacDonald model in the form of Koshino *et al.* [4, 5] as implemented in `pybandstructure` (`koshino`). In the basis of plane waves $\boldsymbol{k} + \boldsymbol{G}$ ($\boldsymbol{G}$ in the moiré reciprocal lattice, $|\boldsymbol{G}| \le G_{\max}|\boldsymbol{G}_M|$ with $G_{\max} = 3$, i.e. 37 plane waves, so that $H(\boldsymbol{k})$ is a 148×148 matrix) and of the four orbitals $(\ell, \sigma) \in \{t, b\} \times \{A, B\}$,

$$H_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k}) = \delta_{\boldsymbol{G}\boldsymbol{G}'} \begin{pmatrix} h_t(\boldsymbol{k}+\boldsymbol{G}) & 0 \\ 0 & h_b(\boldsymbol{k}+\boldsymbol{G}) \end{pmatrix} + \sum_{j=1}^{3} \Big[\delta_{\boldsymbol{G}-\boldsymbol{G}',\boldsymbol{g}_j} \begin{pmatrix} 0 & T_j \\ 0 & 0 \end{pmatrix} + \text{h.c.} \Big],$$
$$h_\ell(\boldsymbol{k}) = \hbar v \left[R_\ell(\boldsymbol{k} - \boldsymbol{K}_\ell) \right] \cdot (\tau\sigma_x, -\sigma_y), \tag{11}$$

where $R_{t,b} = O(\pm\theta_{\mathrm{TBG}}/2)$ rotates the two layers in opposite directions, and $\boldsymbol{K}_{1,2} = \frac{|\boldsymbol{G}_M|}{\sqrt{3}}(-\frac{\sqrt{3}}{2}, \pm\frac{1}{2})$ are the moiré κ points at which the top (1) and bottom (2) Dirac cones sit. The moiré reciprocal lattice is spanned by $\boldsymbol{b}_1, \boldsymbol{b}_2$ with $|\boldsymbol{b}_{1,2}| = |\boldsymbol{G}_M| = 8\pi \sin(\theta_{\mathrm{TBG}}/2)/(\sqrt{3}a)$, $a = 0.246$ nm the graphene lattice constant. The interlayer tunneling connects plane waves that differ by $\boldsymbol{g}_j \in \{0, \boldsymbol{b}_1, \boldsymbol{b}_2\}$ with the 2×2 sublattice matrices of Koshino *et al.* [5]

$$T_1 = \begin{pmatrix} u_{AA} & u_{AB} \\ u_{AB} & u_{AA} \end{pmatrix}, \quad T_2 = \begin{pmatrix} u_{AA} & u_{AB}\varpi^{-\tau} \\ u_{AB}\varpi^{\tau} & u_{AA} \end{pmatrix}, \quad T_3 = \begin{pmatrix} u_{AA} & u_{AB}\varpi^{\tau} \\ u_{AB}\varpi^{-\tau} & u_{AA} \end{pmatrix}, \qquad \varpi = e^{2\pi \mathrm{i}/3}, \tag{12}$$

where u_{AA} and u_{AB} are the tunneling amplitudes between equal and opposite sublattices (the assignment of T_j to $\boldsymbol{g}_j$ follows `pybandstructure.koshino`). Parameters (Ref. [1]): $\theta_{\mathrm{TBG}} = 1.20°$, $u_{AB} = 117$ meV, $u_{AA} = 0.7u_{AB}$ (the reduced value mimics lattice relaxation) and $\hbar v = 0.657$ eV nm. The two central (flat) bands are then 24–25 meV wide and separated from the remote bands by ≈ 44 meV. The code also reports the Bistritzer–MacDonald coupling $\alpha = u_{AB}/(2\hbar v_0 k_D \sin\frac{\theta_{\mathrm{TBG}}}{2})$ with the reference value $\hbar v_0 k_D = 9.905$ eV of Ref. [4] (`TBGParams.alpha`, $\alpha = 0.56$), and the corresponding first-order estimate of the renormalised Dirac velocity [4]

$$v_{\mathrm{BM}} = v\,\frac{1 - 3\alpha^2}{1 + 6\alpha^2} \tag{13}$$

($\hbar v_{\mathrm{BM}} = 0.010$ eV nm, `TBGParams.v_renorm`). Both numbers are used only to set the default broadening η (section 10). With the actual $\hbar v$ the coupling would be $u_{AB}/(\hbar v|\boldsymbol{K}_1 - \boldsymbol{K}_2|) = 0.50$, so v_{BM} is an order of magnitude estimate of the flat-band velocity, not a fitted value. Only the $2n_b = 8$ bands closest to neutrality are computed (`TBGParams.n_bands`$= 4$). The electrostatic environment differs from Ref. [1], which uses an isotropic background $\varepsilon_r = 10$: here the TBG sits in anisotropic hBN, $(\varepsilon_\parallel, \varepsilon_z) = (6.41, 3)$, with vacuum between its layers (`TBGSLGConfig.eps_par`, `eps_z`, `eps_interlayer`), so plasmon energies are not expected to coincide with that reference.

Model frame vs lab frame. The model momentum $\boldsymbol{k}$ is measured from the moiré γ point. The graphene (lab) momentum of the plane wave $\boldsymbol{k} + \boldsymbol{G}$ in either layer is

$$\boldsymbol{p} = \boldsymbol{k} + \boldsymbol{G} + \boldsymbol{\Gamma}^{\mathrm{lab}}, \qquad \boldsymbol{K}^{\mathrm{lab}}_{\mathrm{top}} = \boldsymbol{\Gamma}^{\mathrm{lab}} + \boldsymbol{K}_1, \quad \boldsymbol{K}^{\mathrm{lab}}_{\mathrm{bot}} = \boldsymbol{\Gamma}^{\mathrm{lab}} + \boldsymbol{K}_2, \tag{14}$$

where $\boldsymbol{\Gamma}^{\mathrm{lab}}$ (`gamma_point_tblg`) lies on the x axis and $|\boldsymbol{K}^{\mathrm{lab}}_{\mathrm{top,bot}}| = 4\pi/3a$ at angles $\pm\theta_{\mathrm{TBG}}/2$ (`tbg_model.model_frame`). This is checked in `test_model_frame_dirac_points`, where the decoupled model has a zero-energy top-layer state at $\boldsymbol{K}_1$. This relation is what the QTM kinematics needs (section 7.3).

Bloch states. $H(\boldsymbol{k})u_{n\boldsymbol{k}} = E_{n\boldsymbol{k}}u_{n\boldsymbol{k}}$, where the eigenvector has components $u_{n\boldsymbol{k}}(\boldsymbol{G}, \ell, \sigma)$, stored as `u[4 i_G + 2l + sigma]` with $i_{\boldsymbol{G}}$ the index of the plane wave $\boldsymbol{G}$. The states are stored once (`band_structure.wavefunctions`) and the *same* gauge is used for the form factors, the self-energy and the tunneling amplitudes. The chemical potential is fixed by the filling at temperature $T = 5$ K (charge neutrality by default), and energies are measured from it, $\xi_{n\boldsymbol{k}} = E_{n\boldsymbol{k}} - \mu$.

Layer-resolved form factors. The Coulomb vertices of the screening and of the self-energy are the matrix elements of the density of each layer between Bloch states, $\rho^{\ell}_{nm}(\boldsymbol{k},\boldsymbol{q}) = \langle u_{n\boldsymbol{k}}|P_\ell\,\hat\tau_{\boldsymbol{G}_0}|u_{m,\boldsymbol{k}+\boldsymbol{q}}\rangle$ (`densityL[l, n, m, k, q, G]`), where P_ℓ projects on layer ℓ and $\hat\tau_{\boldsymbol{G}_0}$ compensates the folding of $\boldsymbol{k}+\boldsymbol{q}$ into the first zone. Their derivation from the interaction Hamiltonian, and the reason why the layer index is needed, are given in sections 5.1 and 5.2.

Doping and Hartree. For a doped sample the first-shell Hartree potential $V_H = \sum_{|\boldsymbol{G}|=|\boldsymbol{G}_M|}\frac{2\pi e^2}{\varepsilon_H|\boldsymbol{G}|}\delta n(\boldsymbol{G})e^{i\boldsymbol{G}\cdot\boldsymbol{r}}$, with ε_H the background dielectric constant and $\delta n(\boldsymbol{G})$ the Fourier components of the electron density, can be solved self-consistently (`tbg_model.apply_hartree`, which needs the development version of `pybandstructure`). The densities $\delta n(\boldsymbol{G})$ are measured from charge neutrality (regularised occupations), and G_0 is then built on the Hartree bands. At neutrality the Hartree potential of the BM model vanishes.

3.3 Moiré Brillouin-zone meshes

$\boldsymbol{k}$ and $\boldsymbol{q}$ run over the same mesh of $N_s \times N_s$ points of the moiré Brillouin zone, mapped into its Wigner–Seitz cell (`n_sites` $= N_s$, fig. 8a). Every mesh point represents the same fraction $1/N_s^2$ of the zone, whose area is $A_{\mathrm{BZ}} = \frac{\sqrt{3}}{2}|\boldsymbol{G}_M|^2$. Momentum integrals become sums with the weights $w_{\boldsymbol{q}} = A_{\mathrm{BZ}}/(2\pi N_s)^2$, derived in eq. (49). These weights do not contain the spin–valley degeneracy, which is added explicitly where it belongs (in χ^0, section 5.11). The band structure is computed for valley K, and valley K' follows by time reversal (section 4.1).

4 Screening: polarisability, dielectric matrix and screened interaction

4.1 Independent-particle polarisability

For a subsystem with Bloch states $|n\boldsymbol{k}\rangle$ the layer-resolved Lindhard function [10] is

$$\chi^0_{\ell\ell'}(\boldsymbol{q},\omega) = \frac{g}{A}\sum_{\boldsymbol{k},n,m}\frac{n_{\mathrm{F}}(\xi_{n\boldsymbol{k}}) - n_{\mathrm{F}}(\xi_{m,\boldsymbol{k}+\boldsymbol{q}})}{\omega + i\eta + \xi_{n\boldsymbol{k}} - \xi_{m,\boldsymbol{k}+\boldsymbol{q}}}\,\rho^{\ell}_{nm}(\boldsymbol{k},\boldsymbol{q})\,\left[\rho^{\ell'}_{nm}(\boldsymbol{k},\boldsymbol{q})\right]^* \qquad [\mathrm{nm}^{-2}\,\mathrm{eV}^{-1}], \tag{15}$$

the response of the density of layer ℓ to a potential acting on layer ℓ', written with the form factors of eq. (27), one for each layer index of the bubble (fig. 4c). It is negative at $\omega = 0$ ($\chi^0 \to -D(E_F)$ for $q \to 0$ in a metal), retarded ($\mathrm{Im}\,\chi^0_{\ell\ell} \le 0$ for $\omega > 0$) and, for the head ($\boldsymbol{G} = \boldsymbol{G}' = 0$), a 2×2 matrix for TBG (`LayerHelpers.chi0_block`, `tbg_model.chi0_tbg`). The degeneracy $g = 4$ counts spin and valley. The fast kernel evaluates eq. (15) for all $(\ell, \boldsymbol{G}, \ell', \boldsymbol{G}')$ at once as a matrix product over the transitions with $n_{\mathrm{F}}(\xi_{n\boldsymbol{k}}) \neq n_{\mathrm{F}}(\xi_{m\boldsymbol{k}+\boldsymbol{q}})$, and agrees with `pybandstructure`'s `polarization` to 3×10^{-15}.

For graphene the $T = 0$ bubble is known in closed form [8] (appendix A, `dirac_chi.chi_dirac_phys`). It is written with the split root $\sqrt{z-1}\sqrt{z+1}$, analytic in the upper half plane, and evaluated at complex $\omega + i\eta$. A charge-neutral tip contributes only the interband part $\chi^0 = -\frac{g}{16}\frac{q^2}{\sqrt{(\hbar v_F q)^2 - (\omega+i\eta)^2}}$ (no Drude weight).

Both valleys of TBG. The continuum model is diagonalised for valley K only. Write $X(\boldsymbol{q}) = g\chi^K(\boldsymbol{q})$ for the bubble (15) of that valley multiplied by $g = 4$. Time reversal maps valley K at $\boldsymbol{k}$ onto K' at $-\boldsymbol{k}$, $E_{K'}(\boldsymbol{k}) = E_K(-\boldsymbol{k})$ and $u_{K'}(\boldsymbol{k}) = u_K(-\boldsymbol{k})^*$, and inserted in (15) this gives $\chi^{K'}_{ab}(\boldsymbol{q}) = \chi^K_{ba}(-\boldsymbol{q})$. The physical (spin × valley) response is therefore

$$\chi^0_{\mathrm{TBG}}(\boldsymbol{q},\omega) = \tfrac{g}{2}\left[\chi^K(\boldsymbol{q},\omega) + \chi^K(-\boldsymbol{q},\omega)^T\right] = \tfrac{1}{2}\left[X(\boldsymbol{q},\omega) + X(-\boldsymbol{q},\omega)^T\right] \tag{16}$$

(`tbg_model.chi0_tbg`), as in Ref. [1], where the RPA polarisation sums over both valleys related by time reversal. It satisfies the Onsager relation $\chi^0(\boldsymbol{q}) = \chi^0(-\boldsymbol{q})^T$, hence $W(\boldsymbol{q}) = W(-\boldsymbol{q})^T$, which is

also what allows the frequency integral of the self-energy to be folded onto $\omega > 0$ (section 5.4). Both terms are needed: the bands of a single valley are not inversion symmetric, $E_K(\boldsymbol{k}) \neq E_K(-\boldsymbol{k})$, so $X(\boldsymbol{q})$ alone is not time-reversal symmetric at finite $\boldsymbol{q}$ (its long-wavelength static limit is valley-even). On the few Wigner–Seitz boundary points where $-\boldsymbol{q}$ is stored as $-\boldsymbol{q} + \boldsymbol{G}_0$, the head at the stored representative is used (same $|\boldsymbol{q}|$). The Onsager relation then holds on the whole mesh to round-off (notebook 03, `test_valley_symmetrized_chi0_is_time_reversal_symmetric`).

Block-diagonal χ^0 of a stack. For decoupled subsystems (assumption 2)

$$\chi^0(\boldsymbol{q},\omega) = \begin{pmatrix} \chi^0_{\mathrm{tip}}(|\boldsymbol{q}|,\omega) & 0 \\ 0 & \chi^0_{\mathrm{TBG}}(\boldsymbol{q},\omega) \end{pmatrix} \qquad (3\times 3,\ \texttt{screening.assemble_chi0}). \tag{17}$$

The tip is unstructured on the moiré scale: its response depends on $|\boldsymbol{q}|$ only and has no umklapp.

4.2 Dielectric matrix and screened interaction

The total potential acting on the electrons is the external one plus the Hartree potential of the induced density, $\phi = \phi_{\mathrm{ext}} + V\delta n$ with $\delta n = \chi^0\phi$. Hence $\phi = \varepsilon^{-1}\phi_{\mathrm{ext}}$ with

$$\boxed{\varepsilon(\boldsymbol{q},\omega) = 1 - V(\boldsymbol{q},\omega)\,\chi^0(\boldsymbol{q},\omega)} \qquad \text{and} \qquad \boxed{W(\boldsymbol{q},\omega) = V + V\chi^0 W = \varepsilon^{-1}(\boldsymbol{q},\omega)\,V(\boldsymbol{q},\omega)}, \tag{18}$$

matrices in sheet space (and in $(\ell, \boldsymbol{G})$ space with local fields, $V_{(\ell\boldsymbol{G}),(\ell'\boldsymbol{G}')} = V_{\ell\ell'}(|\boldsymbol{q}+\boldsymbol{G}|)\delta_{\boldsymbol{G}\boldsymbol{G}'}$). The product order matters: V and χ^0 do not commute, so $\varepsilon^{-1}V \neq V\varepsilon^{-1}$. Both V and χ^0 are evaluated at $\omega + \mathrm{i}\eta$ with the *same* η (`screening.screened_interaction`, `tbg_stack.build_stack_eps_W`, `slg_stack.stack_eps_and_V`): an undamped collective mode then appears in $-\operatorname{Im} W$ as a Lorentzian of width $\simeq \eta$ that a finite frequency grid can integrate, instead of a δ function.

4.3 Spectral density, loss function and passivity

W is analytic in the upper half plane and its dynamical part $W - V$ decays at large $|\omega|$. Its spectral representation is

$$W(\boldsymbol{q},z) = V(\boldsymbol{q}) + \int_{-\infty}^{\infty} \mathrm{d}\nu\, \frac{S(\boldsymbol{q},\nu)}{z-\nu}, \qquad S(\boldsymbol{q},\nu) = \frac{\mathrm{i}}{2\pi}\left[W(\boldsymbol{q},\nu+\mathrm{i}0) - W(\boldsymbol{q},\nu+\mathrm{i}0)^\dagger\right], \tag{19}$$

with $S(\boldsymbol{q},-\nu) = -S(-\boldsymbol{q},\nu)^T$. For a passive medium S is Hermitian and positive semidefinite at $\nu > 0$ (it is the absorption of the stack). When W is a symmetric matrix, as for reciprocal systems, $S = -\operatorname{Im} W/\pi$ element by element (`spectral_density_matrix(W, hermitian=False)`). The Hermitian form is used by default because it makes the causality of Σ manifest (section 5.10). For the stacks considered here W is symmetric to $\lesssim 10^{-10}$ and the two coincide. The *loss function* of a set of sheets is $L(\boldsymbol{q},\omega) = -\operatorname{Im}\operatorname{Tr}_{\mathrm{sheets}} W(\boldsymbol{q},\omega) \geq 0$. Its peaks locate collective modes but also inherit every structure of $\operatorname{Im}\chi^0$, so they are a display tool, not a mode criterion.

4.4 Collective modes and plasmon-pole data

Collective modes are zeros of $\det\varepsilon$ [10]: they are located as prominent minima of $|\lambda_{\min}(\varepsilon(\boldsymbol{q},\omega))|$ on the real axis (prominence filter against continuum ripples, `find_plasmon_poles_at_q`). Around each pole $\omega_p(\boldsymbol{q})$, with p labelling the modes found at fixed $\boldsymbol{q}$, the residue *matrix* of W is

$$R_p(\boldsymbol{q}) = \int_{\omega_p-\Delta_-}^{\omega_p+\Delta_+} \mathrm{d}\omega\; S(\boldsymbol{q},\omega), \tag{20}$$

integrated over a window $[\omega_p - \Delta_-, \omega_p + \Delta_+]$ of half-width 5η, truncated at the mid-points between neighbouring poles and, optionally, on a refined local grid (`build_plasmon_pole_data(eps_builder=..., n_refine=129)`). R_p is Hermitian positive semidefinite (PSD). A window of $\pm 5\eta$

captures $\frac{2}{\pi}\arctan 5 \simeq 87\%$ of a Lorentzian. The plasmon-pole approximation (PPA) replaces $S(\boldsymbol{q},\omega) \to \sum_p R_p(\boldsymbol{q})\delta(\omega-\omega_p(\boldsymbol{q}))$ for $\omega>0$. The fraction of $\int S$ captured by the poles measures its validity. In doped graphene above $q \simeq k_F$ and in TBG at small q the modes are Landau damped and most of the weight is in the continuum, so the PPA is used only as a diagnostic and the production route is the full-frequency one.

4.5 Layer-symmetry channels

For a pair of sheets (a,b) write the 2×2 block of W with $P_\pm = u_\pm u_\pm^T$, $u_\pm = (1,\pm 1)/\sqrt{2}$:

$$W_{\text{block}} = \underbrace{P_+ W P_+}_{\text{in phase}} + \underbrace{P_- W P_-}_{\text{out of phase}} + \underbrace{P_+ W P_- + P_- W P_+}_{\text{cross}}. \tag{21}$$

This is an identity, exact and additive, and since Σ is linear in S the split carries over to the self-energy (`screening.split_layer_channels`, `slg_self_energy.sigma_full_frequency_channels_kernel`). For two equivalent sheets $u_\pm$ are exact eigenvectors of W and the cross term vanishes, and the in-phase and out-of-phase parts are the optical and acoustic plasmon channels. It is a symmetry projection, valid also where the modes are damped and are not poles. Whether the out-of-phase part is a second mode or the same mode re-weighted by geometry must be checked from the peak positions of the two channels (with an inert tip both peak at the same ω).

5 The G_0W self-energy

This section derives the self-energy that the package evaluates, eq. (43) below, starting from the Coulomb Hamiltonian of the stack. The derivation is standard many-body perturbation theory [29, 14, 9, 11, 17]. What is specific to a van der Waals stack is that the interaction is a matrix in the sheet index, and that the electrons of a multilayer sample (the two layers of TBG) are coherent superpositions of states living on different sheets. These two facts are why the layer-resolved density matrix elements appear in every vertex. Figure 3 shows the sequence of steps and where each one is done.

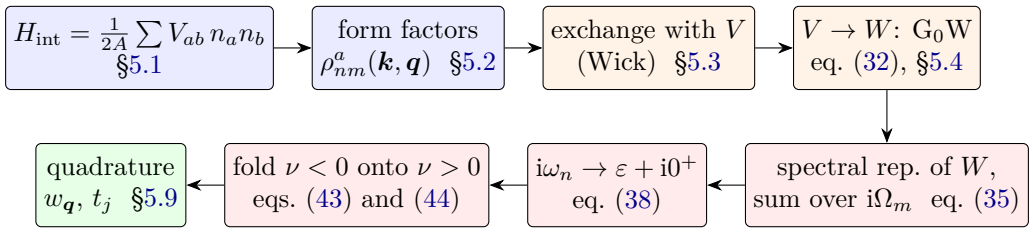


Figure 3: Steps of the derivation of the self-energy. Blue: the interaction and its matrix elements; orange: the diagram; red: the frequency structure; green: the numerical evaluation.

5.1 Interaction Hamiltonian of the stack

Let $\psi_{a\sigma}(\boldsymbol{r})$ be the field operator of an electron on sheet a, sublattice σ, at in-plane position $\boldsymbol{r}$, and $n_a(\boldsymbol{r}) = \sum_\sigma \psi^\dagger_{a\sigma}(\boldsymbol{r})\psi_{a\sigma}(\boldsymbol{r})$ the number density of sheet a. Spin and valley indices are suppressed: the interaction is diagonal in both (section 5.11), and the density operators below include the sum over them. Because the sheets are infinitely thin (assumption 1), the charge density of the stack is $-e\sum_a n_a(\boldsymbol{r})\delta(z-z_a)$, and the electrostatic energy is

$$H_{\text{int}} = \frac{1}{2}\sum_{ab}\int \mathrm{d}^2r\,\mathrm{d}^2r'\; n_a(\boldsymbol{r})\, U_{ab}(\boldsymbol{r}-\boldsymbol{r}')\, n_b(\boldsymbol{r}'), \tag{22}$$

where $U_{ab}(\boldsymbol{r}-\boldsymbol{r}')$ is the energy of an electron at $(\boldsymbol{r},z_a)$ in the potential of a second electron at $(\boldsymbol{r}',z_b)$, including all the image charges induced in the dielectrics and in the gates. It depends only

on $\boldsymbol{r}-\boldsymbol{r}'$ because the stack is homogeneous in the plane, and its Fourier transform is precisely the kernel of section 2, $U_{ab}(\boldsymbol{r}) = A^{-1}\sum_{\boldsymbol{q}} V_{ab}(q)e^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, where A is the normalisation area of the sample. With $n_a(\boldsymbol{q}) = \int \mathrm{d}^2r\, e^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} n_a(\boldsymbol{r})$,

$$H_{\mathrm{int}} = \frac{1}{2A}\sum_{\boldsymbol{q}\neq 0}\sum_{ab} V_{ab}(q)\ :n_a(\boldsymbol{q})\, n_b(-\boldsymbol{q}):\,. \tag{23}$$

The colons denote normal ordering, which removes the interaction of each electron with itself, and the $\boldsymbol{q}=0$ term is cancelled by the neutralising background of the gates and dopants. The density operator obeys $n_a(\boldsymbol{q})^\dagger = n_a(-\boldsymbol{q})$.

Equation (23) is the reason for the layer index. $V_{ab}(q)$ is a genuine matrix: two electrons on the same sheet interact through V_{aa}, two electrons on sheets a distance d apart through $V_{ab}\simeq V_{aa}e^{-\kappa d}$ (eq. (7)), and a gate or a dielectric interface modifies the two differently. The interaction energy of a density fluctuation therefore depends on how its charge is distributed among the sheets, and the operator that couples to the Coulomb field of sheet a is the density of that sheet, $n_a(\boldsymbol{q})$, not the total density. In a monolayer sample this is trivial, since only one sheet is involved, but in TBG a Bloch electron has amplitude on both layers and a transition between two Bloch states moves charge on both, with different weights. These weights are the form factors.

5.2 Layer-resolved density operator in the band basis

In the continuum model of section 3.2 the field operator is expanded in plane waves of the moiré reciprocal lattice,

$$\psi_{a\sigma}(\boldsymbol{r}) = \frac{1}{\sqrt{A}}\sum_{\boldsymbol{k}\in\mathrm{BZ}}\sum_{\boldsymbol{G}} e^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}\, c_{\boldsymbol{k}+\boldsymbol{G},a\sigma}, \qquad c_{\boldsymbol{k}+\boldsymbol{G},a\sigma} = \sum_n u_{n\boldsymbol{k}}(\boldsymbol{G},a,\sigma)\, c_{n\boldsymbol{k}}, \tag{24}$$

where $c_{n\boldsymbol{k}}$ annihilates an electron in the Bloch eigenstate n of $H(\boldsymbol{k})$ and $u_{n\boldsymbol{k}}(\boldsymbol{G},a,\sigma)$ is the component of that eigenvector on plane wave $\boldsymbol{G}$, sheet a, sublattice σ (stored as `u[4 i_G + 2a + sigma]`, $i_{\boldsymbol{G}}$ the index of $\boldsymbol{G}$). The momenta of both layers are measured from the same origin $\boldsymbol{\Gamma}^{\mathrm{lab}}$ (eq. (14)), so this common offset cancels in $\psi^\dagger\psi$. Inserting eq. (24) into the definition of $n_a(\boldsymbol{q})$ and using $\int \mathrm{d}^2r\, e^{\mathrm{i}(\boldsymbol{p}'-\boldsymbol{p})\cdot\boldsymbol{r}} = A\,\delta_{\boldsymbol{p}\boldsymbol{p}'}$:

$$n_a(\boldsymbol{q}) = \sum_{\boldsymbol{k}\in\mathrm{BZ}}\sum_{\boldsymbol{G}\sigma} c^\dagger_{\boldsymbol{k}+\boldsymbol{G},a\sigma}\, c_{\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{G},a\sigma}. \tag{25}$$

If $\boldsymbol{k}+\boldsymbol{q}$ lies outside the first zone we write $\boldsymbol{k}+\boldsymbol{q} = \boldsymbol{k}'+\boldsymbol{G}_0$ with $\boldsymbol{k}'$ in the zone (fig. 8a), so that $c_{\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{G},a\sigma} = c_{\boldsymbol{k}'+(\boldsymbol{G}+\boldsymbol{G}_0),a\sigma}$. Expanding both operators in Bloch states gives

$$n_a(\boldsymbol{q}) = \sum_{\boldsymbol{k}}\sum_{nm} \rho^a_{nm}(\boldsymbol{k},\boldsymbol{q})\, c^\dagger_{n\boldsymbol{k}}\, c_{m,\boldsymbol{k}+\boldsymbol{q}}, \tag{26}$$

with the *form factors*

$$\rho^a_{nm}(\boldsymbol{k},\boldsymbol{q}) = \sum_{\boldsymbol{G}\sigma} u^*_{n\boldsymbol{k}}(\boldsymbol{G},a,\sigma)\, u_{m\boldsymbol{k}'}(\boldsymbol{G}+\boldsymbol{G}_0,a,\sigma) = \langle u_{n\boldsymbol{k}}|P_a\,\hat{\tau}_{\boldsymbol{G}_0}|u_{m,\boldsymbol{k}'}\rangle, \tag{27}$$

with $c_{m,\boldsymbol{k}+\boldsymbol{q}} \equiv c_{m\boldsymbol{k}'}$. Here P_a projects on the two sublattices of sheet a and $\hat{\tau}_{\boldsymbol{G}_0}$ shifts the plane-wave index by $\boldsymbol{G}_0$ (`densityL[a, n, m, k, q, 0]` in `LayerHelpers`; the mesh index of $\boldsymbol{k}'$ is `k_plus_q_ind[q, k]`). The form factor $\rho^a_{nm}(\boldsymbol{k},\boldsymbol{q})$ is the amplitude for the transition $m,\boldsymbol{k}+\boldsymbol{q}\to n,\boldsymbol{k}$ to be accompanied by a density change of wavevector $\boldsymbol{q}$ on sheet a. Three properties are used below. Since $n_a(\boldsymbol{q})^\dagger = n_a(-\boldsymbol{q})$,

$$n_a(-\boldsymbol{q}) = \sum_{\boldsymbol{k}}\sum_{nm}\left[\rho^a_{nm}(\boldsymbol{k},\boldsymbol{q})\right]^* c^\dagger_{m,\boldsymbol{k}+\boldsymbol{q}}\, c_{n\boldsymbol{k}}. \tag{28}$$

Summing over the sheets of the sample gives the total-density matrix element, which at $\boldsymbol{q}=0$ is the overlap of two eigenvectors of the same Hamiltonian, $\sum_a \rho^a_{nm}(\boldsymbol{k},0) = \delta_{nm}$ (`test_density_`

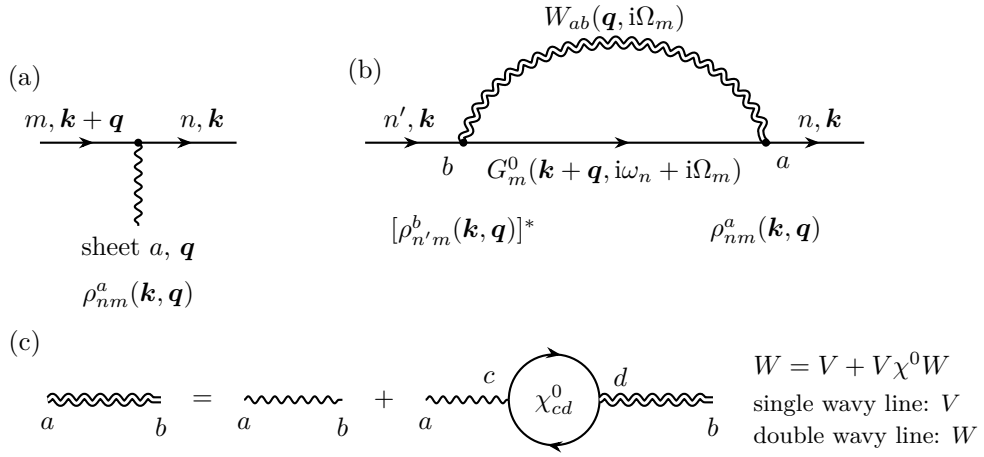


Figure 4: Diagrams of the G_0W self-energy. (a) Coulomb vertex: an electron scatters from $m, \boldsymbol{k}+\boldsymbol{q}$ to $n, \boldsymbol{k}$ and the density of sheet a changes by a Fourier component $\boldsymbol{q}$, with amplitude given by the form factor (27). (b) The self-energy (32): the external electron emits a density fluctuation on sheet b, propagates as m, and reabsorbs the fluctuation on sheet a. (c) The screened interaction W_{ab} is the bare kernel V_{ab} dressed by the polarisation bubbles of all sheets of the stack (eq. (18)).

`matrix_completeness`). Finally, sample Bloch states have no amplitude on the tip, because tip and sample are not tunnel coupled (assumption 2): $\rho^{\mathrm{tip}}_{nm} = 0$ for sample states, so only the sample sheets appear in the vertices of the sample self-energy, while the tip enters through the screening. For a momentum transfer $\boldsymbol{q}+\boldsymbol{G}$ that differs from $\boldsymbol{q}$ by a moiré vector, the same construction gives the umklapp form factors $\rho^a_{nm}(\boldsymbol{k},\boldsymbol{q};\boldsymbol{G})$ (last index of `densityL`). Only $\boldsymbol{G}=0$ is kept in the self-energy (assumption 4). Physically this restricts the momentum transferred to the electron to the first moiré zone.

For graphene there is one sheet, the plane-wave sum reduces to a single term, and the form factor is the spinor overlap $\rho_{ss'}(\boldsymbol{k},\boldsymbol{q}) = \langle u^s_{\boldsymbol{k}} | u^{s'}_{\boldsymbol{k}+\boldsymbol{q}} \rangle$, whose square is the chirality factor of eq. (10).

5.3 The exchange term: origin of the vertex structure

The structure of the self-energy is already visible at first order in V. Insert eqs. (27) and (28) in eq. (23):

$$H_{\mathrm{int}} = \frac{1}{2A}\sum_{\boldsymbol{q}}\sum_{ab} V_{ab}(q) \sum_{\boldsymbol{k}\boldsymbol{k}'}\sum_{nmn'm'} \rho^a_{nm}(\boldsymbol{k},\boldsymbol{q})\left[\rho^b_{n'm'}(\boldsymbol{k}',\boldsymbol{q})\right]^* c^\dagger_{n\boldsymbol{k}}\, c^\dagger_{m',\boldsymbol{k}'+\boldsymbol{q}}\, c_{n'\boldsymbol{k}'}\, c_{m,\boldsymbol{k}+\boldsymbol{q}}, \tag{29}$$

where normal ordering has moved $c^\dagger_{m',\boldsymbol{k}'+\boldsymbol{q}}$ to the left at the price of a sign, compensated by reversing the order of the two annihilation operators. The first-order self-energy follows from Wick's theorem, i.e. from the mean-field decoupling of the four-operator product in the non-interacting ground state, where $\langle c^\dagger_{m'\boldsymbol{p}'} c_{m\boldsymbol{p}}\rangle = \delta_{mm'}\delta_{\boldsymbol{p}\boldsymbol{p}'} n_{\mathrm{F}}(\xi_{m\boldsymbol{p}})$. The *exchange* contraction pairs $c^\dagger_{m',\boldsymbol{k}'+\boldsymbol{q}}$ with $c_{m,\boldsymbol{k}+\boldsymbol{q}}$. It requires $\boldsymbol{k}' = \boldsymbol{k}$ and $m' = m$, and moving $c^\dagger_{m',\boldsymbol{k}'+\boldsymbol{q}}$ next to $c_{m,\boldsymbol{k}+\boldsymbol{q}}$ costs one more sign. The other exchange contraction, of $c^\dagger_{n\boldsymbol{k}}$ with $c_{n'\boldsymbol{k}'}$, gives the same operator after relabelling $\boldsymbol{q}\to-\boldsymbol{q}$, $a\leftrightarrow b$, and doubles the result, which cancels the factor $\frac{1}{2}$. The two *Hartree* contractions, inside each density, set $\boldsymbol{q}=0$, which is excluded. The result is a one-body operator $\sum_{\boldsymbol{k}}\sum_{nn'}\Sigma^x_{nn'}(\boldsymbol{k})\, c^\dagger_{n\boldsymbol{k}} c_{n'\boldsymbol{k}}$ with

$$\Sigma^x_{nn'}(\boldsymbol{k}) = -\frac{1}{A}\sum_{\boldsymbol{q}}\sum_{m}\sum_{ab} \rho^a_{nm}(\boldsymbol{k},\boldsymbol{q})\, V_{ab}(q)\, \left[\rho^b_{n'm}(\boldsymbol{k},\boldsymbol{q})\right]^* n_{\mathrm{F}}(\xi_{m,\boldsymbol{k}+\boldsymbol{q}}). \tag{30}$$

This is the Fock self-energy. It contains the three ingredients of the G_0W formula: the vertex ρ^a_{nm} that takes $n,\boldsymbol{k}$ to $m,\boldsymbol{k}+\boldsymbol{q}$ while creating a density fluctuation on sheet a, the interaction V_{ab} that carries this fluctuation to sheet b, and the conjugate vertex $[\rho^b_{n'm}]^*$ that brings the electron back to $n',\boldsymbol{k}$. The overall minus sign is the fermionic exchange sign, and the occupation $n_{\mathrm{F}}(\xi_m)$ states that

only occupied intermediate states contribute to exchange. When $n \neq n'$ the result is a matrix in band space, because the electron can leave from band n' and return to band n through the same intermediate state.

5.4 Dynamical screening and the G_0W approximation

The exchange term uses the bare, instantaneous interaction. In the stack, the density fluctuation created at the first vertex polarises all the other electrons, of the sample and of the tip, and the field that reaches the second vertex is screened and retarded. Hedin's equations [29] express the exact self-energy as $\Sigma = \mathrm{i}GW\Gamma$, with $W = V + VPW$ the screened interaction, P the irreducible polarisability and Γ the vertex function. The GW approximation sets $\Gamma = 1$, so that $P = -\mathrm{i}GG$ is the RPA bubble and $\Sigma = \mathrm{i}GW$ is the Fock diagram with V replaced by W (fig. 4b) [29, 15, 17, 18]. In the one-shot G_0W variant used here, G is the non-interacting propagator of the sample and W is built from the non-interacting polarisability χ^0 (assumption 3).

Screened interaction in sheet space. The layer-resolved W is the bare interaction dressed by the density response of the stack,

$$W_{ab}(\boldsymbol{q}, \mathrm{i}\Omega_m) = V_{ab}(q) + \sum_{cd} V_{ac}(q)\, \chi_{cd}(\boldsymbol{q}, \mathrm{i}\Omega_m)\, V_{db}(q), \tag{31}$$

$$\chi_{cd}(\boldsymbol{q}, \mathrm{i}\Omega_m) = -\frac{1}{A}\int_0^\beta \mathrm{d}\tau\, e^{\mathrm{i}\Omega_m\tau} \langle T_\tau\, n_c(\boldsymbol{q}, \tau)\, n_d(-\boldsymbol{q}, 0)\rangle,$$

where χ_{cd} is the response of the density of sheet c to a potential applied to sheet d, $\beta = 1/k_BT$ and T_τ orders in imaginary time. It is the effective interaction between a density fluctuation of wavevector $\boldsymbol{q}$ on sheet a and one on sheet b, mediated by every polarisable sheet of the junction. In the RPA $\chi = \chi^0 + \chi^0 V \chi$, which gives $W = V + V\chi^0 W$ (fig. 4c) and therefore the matrix relation $W = \varepsilon^{-1}V$ of eq. (18). The Lindhard function χ^0_{cd} of eq. (15) is built with the same form factors, two per bubble, one for each sheet index.

Matsubara form. The Feynman rules of the imaginary-time formalism [9, 11, 12] assign to diagram 4b a propagator $G^0_m(\boldsymbol{k}+\boldsymbol{q}, \mathrm{i}\omega_n + \mathrm{i}\Omega_m)$ for the internal line, the interaction $W_{ab}(\boldsymbol{q}, \mathrm{i}\Omega_m)$, a vertex factor at each end, a factor $1/(\beta A)$ for the free sums over $\boldsymbol{q}$ and over the bosonic frequency, and the exchange sign. The result is

$$\Sigma_{nn'}(\boldsymbol{k}, \mathrm{i}\omega_n) = -\frac{1}{\beta A}\sum_{\boldsymbol{q}}\sum_{\mathrm{i}\Omega_m}\sum_{m}\sum_{ab} \rho^a_{nm}(\boldsymbol{k}, \boldsymbol{q})\, W_{ab}(\boldsymbol{q}, \mathrm{i}\Omega_m)\, \left[\rho^b_{n'm}(\boldsymbol{k}, \boldsymbol{q})\right]^*\, G^0_m(\boldsymbol{k}+\boldsymbol{q}, \mathrm{i}\omega_n + \mathrm{i}\Omega_m), \tag{32}$$

with the fermionic and bosonic Matsubara frequencies $\omega_n = (2n+1)\pi/\beta$ and $\Omega_m = 2m\pi/\beta$, and the non-interacting propagator $G^0_m(\boldsymbol{p}, \mathrm{i}\omega_n) = 1/(\mathrm{i}\omega_n - \xi_{m\boldsymbol{p}})$. Energy is conserved at the vertices, which is why the bosonic frequency Ω_m carried by W appears in the internal electron line. The indices a, b run over the sheets of the sample, m over the bands included in the calculation, and $\frac{1}{A}\sum_{\boldsymbol{q}} \to \int \mathrm{d}^2q/(2\pi)^2$ in the thermodynamic limit.

The sign and the prefactor of eq. (32) can be checked against eq. (30). For a static interaction, $W \to V$, the frequency sum involves only the propagator. With the convergence factor $e^{\mathrm{i}\Omega_m 0^+}$ that accompanies an equal-time interaction [9],

$$\frac{1}{\beta}\sum_{\mathrm{i}\Omega_m} \frac{e^{\mathrm{i}\Omega_m 0^+}}{\mathrm{i}\Omega_m - (\xi_m - \mathrm{i}\omega_n)} = -n_{\mathrm{B}}(\xi_m - \mathrm{i}\omega_n) = n_{\mathrm{F}}(\xi_m), \tag{33}$$

where the last step uses $e^{-\mathrm{i}\beta\omega_n} = -1$. Equation (32) then reduces to eq. (30), as it must.

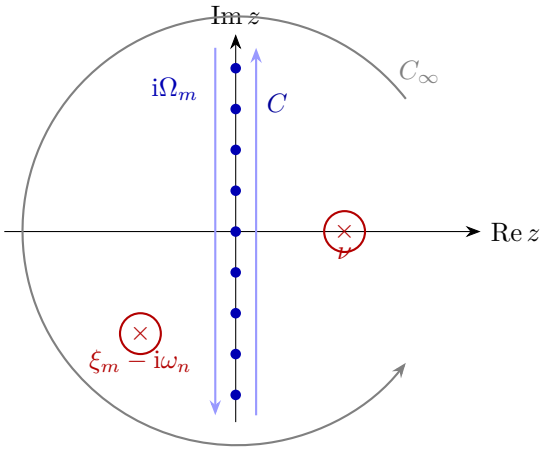


Figure 5: Contour for the bosonic sum (35). The Bose function $n_\mathrm{B}(z)$ has simple poles of residue $1/\beta$ at $z = \mathrm{i}\Omega_m$ (blue dots), so the sum is $\oint_C \frac{\mathrm{d}z}{2\pi\mathrm{i}} n_\mathrm{B}(z)F(z)$ along C. Deforming C into the large circle C_∞, where the integrand vanishes, leaves the clockwise circles around the two poles of F (red crosses).

5.5 Spectral representation of W and the frequency sum

Split the interaction into its bare and dynamical parts, $W = V + W_c$. The bare part gives the exchange term (30), which is not included (section 5.12). The dynamical part is analytic in the complex frequency plane away from the real axis and vanishes for $|z| \to \infty$, because the polarisability does. For a finite set of bands $\chi^0(\boldsymbol{q}, z) \propto z^{-2}$ (the z^{-1} term is the expectation value of a commutator of densities, which vanishes, and the z^{-2} term is fixed by the f-sum rule [10]), while for the unbounded Dirac cone $\chi^0 \propto z^{-1}$. It therefore obeys the spectral representation (19),

$$W_c(\boldsymbol{q}, z) = \int_{-\infty}^{\infty} \mathrm{d}\nu \, \frac{S(\boldsymbol{q}, \nu)}{z - \nu}, \qquad S(\boldsymbol{q}, \nu) = \frac{\mathrm{i}}{2\pi}[W^\mathrm{R}(\boldsymbol{q}, \nu) - W^\mathrm{R}(\boldsymbol{q}, \nu)^\dagger], \tag{34}$$

which gives the retarded function for $\operatorname{Im} z > 0$, the advanced one for $\operatorname{Im} z < 0$ (using $W^\mathrm{A} = (W^\mathrm{R})^\dagger$), and the Matsubara function at $z = \mathrm{i}\Omega_m$. Inserted in eq. (32), the only dependence on $\mathrm{i}\Omega_m$ is in the product of two simple poles, and the frequency sum is

$$\frac{1}{\beta} \sum_{\mathrm{i}\Omega_m} \frac{1}{(\mathrm{i}\Omega_m - \nu)(\mathrm{i}\Omega_m + \mathrm{i}\omega_n - \xi_m)} = -\frac{n_\mathrm{B}(\nu) + n_\mathrm{F}(\xi_m)}{\mathrm{i}\omega_n + \nu - \xi_m}. \tag{35}$$

The derivation is the standard contour argument [9, 11] (fig. 5). Write $F(z) = 1/[(z - \nu)(z + \mathrm{i}\omega_n - \xi_m)]$. The function $n_\mathrm{B}(z) = 1/(e^{\beta z} - 1)$ has simple poles with residue $1/\beta$ at $z = \mathrm{i}\Omega_m$, hence $\frac{1}{\beta}\sum_m F(\mathrm{i}\Omega_m) = \oint_C \frac{\mathrm{d}z}{2\pi\mathrm{i}} n_\mathrm{B}(z)F(z)$ along a contour C that encloses the imaginary axis counter-clockwise. Since $F \sim z^{-2}$ the contour can be opened onto a circle at infinity, which contributes nothing, and what remains are the two poles of F, encircled clockwise:

$$\frac{1}{\beta} \sum_{\mathrm{i}\Omega_m} F(\mathrm{i}\Omega_m) = -\frac{n_\mathrm{B}(\nu)}{\nu + \mathrm{i}\omega_n - \xi_m} - \frac{n_\mathrm{B}(\xi_m - \mathrm{i}\omega_n)}{\xi_m - \mathrm{i}\omega_n - \nu}. \tag{36}$$

Because ω_n is a fermionic frequency, $n_\mathrm{B}(\xi_m - \mathrm{i}\omega_n) = 1/(-e^{\beta\xi_m} - 1) = -n_\mathrm{F}(\xi_m)$, and the two terms combine into eq. (35). Using it in eq. (32) with $W \to W_c$, the two minus signs cancel:

$$\Sigma^c_{nn'}(\boldsymbol{k}, \mathrm{i}\omega_n) = \frac{1}{A} \sum_{\boldsymbol{q}} \sum_m \sum_{ab} \rho^a_{nm}(\boldsymbol{k}, \boldsymbol{q}) [\rho^b_{n'm}(\boldsymbol{k}, \boldsymbol{q})]^* \int_{-\infty}^{\infty} \mathrm{d}\nu \; S_{ab}(\boldsymbol{q}, \nu) \, \frac{n_\mathrm{B}(\nu) + n_\mathrm{F}(\xi_m)}{\mathrm{i}\omega_n + \nu - \xi_m}, \tag{37}$$

with the shorthand $\xi_m \equiv \xi_{m,\boldsymbol{k}+\boldsymbol{q}}$ for the energy of the intermediate state.

5.6 Analytic continuation to real frequency

The Matsubara self-energy now depends on $\mathrm{i}\omega_n$ only through the denominator, since the factor $e^{-\mathrm{i}\beta\omega_n} = -1$ has already been used. It is the value at $z = \mathrm{i}\omega_n$ of a function analytic off the real axis and decaying at infinity, and this continuation is unique [13]. The retarded self-energy, which is the one that enters the Dyson equation for the spectral function, is obtained with $\mathrm{i}\omega_n \to \varepsilon + \mathrm{i}0^+$:

$$\Sigma^c_{nn'}(\boldsymbol{k},\varepsilon) = \int \frac{\mathrm{d}^2q}{(2\pi)^2} \sum_m \sum_{ab} \rho^a_{nm}(\boldsymbol{k},\boldsymbol{q}) \left[\rho^b_{n'm}(\boldsymbol{k},\boldsymbol{q})\right]^* \int_{-\infty}^{\infty} \mathrm{d}\nu \; S_{ab}(\boldsymbol{q},\nu) \, \frac{n_\mathrm{B}(\nu) + n_\mathrm{F}(\xi_m)}{\varepsilon + \nu - \xi_m + \mathrm{i}0^+}. \tag{38}$$

This is Eqs. (37) and (39) of Ref. [1], with the sheet index in place of the moiré $\boldsymbol{G}$ index. In eq. (38) ν runs over the whole real axis, and the spectral density at negative ν is not stored by the code. The next step shows that it is not needed.

5.7 Folding onto positive frequencies

Symmetry of S. Two properties of the response of the stack fix S at $\nu < 0$. First, the density $n_a(\boldsymbol{r})$ is a Hermitian operator, so the retarded response to a real potential is real in real space and time. In Fourier space this reads $W^\mathrm{R}_{ab}(\boldsymbol{q},-\nu) = [W^\mathrm{R}_{ab}(-\boldsymbol{q},\nu)]^*$, and inserting it in eq. (34),

$$S(\boldsymbol{q},-\nu) = \frac{\mathrm{i}}{2\pi}\Big[W^\mathrm{R}(-\boldsymbol{q},\nu)^* - W^\mathrm{R}(-\boldsymbol{q},\nu)^T\Big] = -\Big\{\frac{\mathrm{i}}{2\pi}\big[W^\mathrm{R}(-\boldsymbol{q},\nu) - W^\mathrm{R}(-\boldsymbol{q},\nu)^\dagger\big]\Big\}^T = -S(-\boldsymbol{q},\nu)^T. \tag{39}$$

Second, for a time-reversal-symmetric sample the Onsager relation $\chi^0(-\boldsymbol{q},\nu) = \chi^0(\boldsymbol{q},\nu)^T$ holds [10] (for TBG it requires both valleys, eq. (16)). Since V is symmetric, $W(-\boldsymbol{q},\nu) = W(\boldsymbol{q},\nu)^T$ and $S(-\boldsymbol{q},\nu)^T = S(\boldsymbol{q},\nu)$. Together,

$$S(\boldsymbol{q},-\nu) = -S(\boldsymbol{q},\nu)\,: \tag{40}$$

the spectral density is odd in frequency, positive semidefinite at $\nu > 0$ (absorption by the stack) and negative semidefinite at $\nu < 0$.

Folding. Split the ν integral of eq. (38) at $\nu = 0$ and substitute $\nu = -\omega$ in the negative half. With eq. (40), $n_\mathrm{B}(-\omega) = -1 - n_\mathrm{B}(\omega)$ and $1 - n_\mathrm{F}(\xi) = n_\mathrm{F}(-\xi)$,

$$\begin{aligned}\int_{-\infty}^{0} \mathrm{d}\nu \, S(\boldsymbol{q},\nu) \frac{n_\mathrm{B}(\nu) + n_\mathrm{F}(\xi_m)}{\varepsilon + \nu - \xi_m + \mathrm{i}0^+} &= \int_0^\infty \mathrm{d}\omega \, \big[-S(\boldsymbol{q},\omega)\big] \frac{-1 - n_\mathrm{B}(\omega) + n_\mathrm{F}(\xi_m)}{\varepsilon - \omega - \xi_m + \mathrm{i}0^+} \\ &= \int_0^\infty \mathrm{d}\omega \, S(\boldsymbol{q},\omega) \frac{n_\mathrm{B}(\omega) + n_\mathrm{F}(-\xi_m)}{\varepsilon - \omega - \xi_m + \mathrm{i}0^+}.\end{aligned} \tag{41}$$

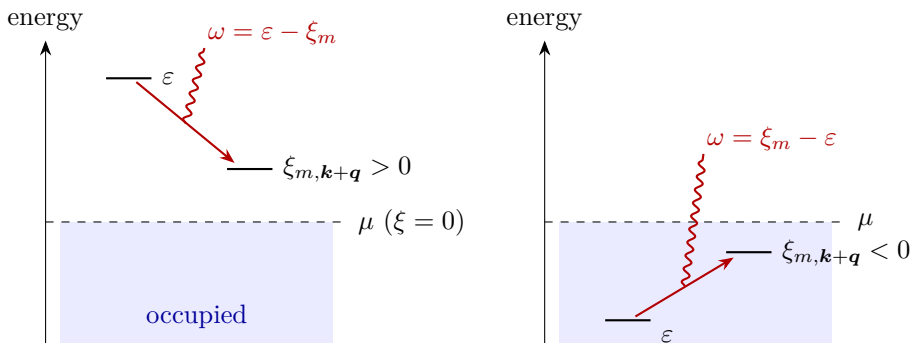
Adding the positive half, which is unchanged,

$$\begin{aligned}\Sigma^c_{nn'}(\boldsymbol{k},\varepsilon) = \int \frac{\mathrm{d}^2q}{(2\pi)^2} \sum_m \sum_{ab} \rho^a_{nm} \big[\rho^b_{n'm}\big]^* \\ \times \int_0^\infty \mathrm{d}\omega \; S_{ab}(\boldsymbol{q},\omega) \left[\frac{n_\mathrm{B}(\omega) + n_\mathrm{F}(-\xi_m)}{\varepsilon - \xi_m - \omega + \mathrm{i}0^+} + \frac{n_\mathrm{B}(\omega) + n_\mathrm{F}(\xi_m)}{\varepsilon - \xi_m + \omega + \mathrm{i}0^+}\right].\end{aligned} \tag{42}$$

The code writes the bracket with the opposite sign of the denominators, which makes the sign of $\mathrm{Im}\,\Sigma$ manifest. Replacing the integral over $\boldsymbol{q}$ by the quadrature of section 5.9, the infinitesimal 0^+ by a small but finite η_Σ, and the upper limit by the largest frequency ω_max of the grid on which S is known,

$$\boxed{\Sigma_{nn'}(\boldsymbol{k},\varepsilon) = -\sum_{\boldsymbol{q}} w_{\boldsymbol{q}} \sum_m \sum_{ab} \rho^a_{nm}(\boldsymbol{k},\boldsymbol{q}) \big[\rho^b_{n'm}(\boldsymbol{k},\boldsymbol{q})\big]^* \int_0^{\omega_\mathrm{max}} \mathrm{d}\omega \; S_{ab}(\boldsymbol{q},\omega) \, \mathcal{D}_m(\omega,\varepsilon)} \tag{43}$$

$$\mathcal{D}_m(\omega,\varepsilon) = \frac{N^+_m(\omega)}{\omega + \xi_m - \varepsilon - \mathrm{i}\eta_\Sigma} + \frac{N^-_m(\omega)}{-\omega + \xi_m - \varepsilon - \mathrm{i}\eta_\Sigma}, \tag{44}$$



(a) electron, $\Theta(\xi_m)$ term (b) hole, $\Theta(-\xi_m)$ term

Figure 6: The two terms of $\mathcal{D}_m$ at $T = 0$. (a) An electron added at $\varepsilon > 0$ relaxes into an empty state m and emits a boson of energy $\varepsilon - \xi_m$. (b) A hole added at $\varepsilon < 0$ relaxes into an occupied state (the hole moves up) and emits a boson of energy $\xi_m - \varepsilon$. The bosonic spectral weight at that energy, $S(\boldsymbol{q}, \omega)$, sets the decay rate.

$$N_m^{\pm}(\omega) = n_{\mathrm{B}}(\omega) + n_{\mathrm{F}}(\mp\xi_m) \xrightarrow{T=0} \Theta(\pm\xi_m). \tag{45}$$

Here $w_{\boldsymbol{q}}$ is the weight of the momentum quadrature (section 5.9), $\xi_m = \xi_{m,\boldsymbol{k}+\boldsymbol{q}}$ is the energy of the intermediate state measured from the chemical potential, $N_m^{\pm}$ are its population factors, and the label Σ stands for the correlation part Σ^c from now on. Equation (43) is implemented in `compute_self_energy_band_matrix` (TBG, band matrix, direct frequency contraction) and in its fast version `compute_self_energy_band_matrix_kernel` (section 5.14). The diagonal elements alone are computed by `compute_self_energy_full_frequency_pybandstructure`, which returns the summand of every pair (a, b) separately so that their sum is Σ_{nn}. For graphene the same formula is evaluated by `slg_self_energy.sigma_full_frequency_fast` and `sigma_full_frequency_kernel` on the radial grid of section 5.9.

5.8 Physical content

At $T = 0$ the Bose factor vanishes for $\omega > 0$ and the populations become step functions. Equation (42) then reads

$$\Sigma_{nn'}(\boldsymbol{k}, \varepsilon) = \int \frac{\mathrm{d}^2 q}{(2\pi)^2} \sum_m \sum_{ab} \rho^a_{nm} [\rho^b_{n'm}]^* \int_0^\infty \mathrm{d}\omega \, S_{ab}(\boldsymbol{q}, \omega) \left[\frac{\Theta(\xi_m)}{\varepsilon - \xi_m - \omega + \mathrm{i}0^+} + \frac{\Theta(-\xi_m)}{\varepsilon - \xi_m + \omega + \mathrm{i}0^+} \right], \tag{46}$$

the form familiar from GW calculations in solids [15, 17]. Its anti-Hermitian part follows from $\mathrm{Im}(x + \mathrm{i}0^+)^{-1} = -\pi\delta(x)$. For the diagonal element,

$$-\operatorname{Im}\Sigma_{nn}(\boldsymbol{k}, \varepsilon) = \pi \int \frac{\mathrm{d}^2 q}{(2\pi)^2} \sum_m \Big[\Theta(\xi_m)\,\Theta(\varepsilon - \xi_m)\,\mathcal{S}_{nm}(\boldsymbol{q}, \varepsilon - \xi_m) + \Theta(-\xi_m)\,\Theta(\xi_m - \varepsilon)\,\mathcal{S}_{nm}(\boldsymbol{q}, \xi_m - \varepsilon) \Big], \tag{47}$$

with $\mathcal{S}_{nm}(\boldsymbol{q}, \omega) = \sum_{ab} \rho^a_{nm} S_{ab}(\boldsymbol{q}, \omega) [\rho^b_{nm}]^* \geq 0$, the spectral density of W seen through the vertex of the transition $n \to m$. This is Fermi's golden rule (fig. 6). An electron added at energy $\varepsilon > 0$ decays into an empty state $m, \boldsymbol{k} + \boldsymbol{q}$ of lower energy $\xi_m \in (0, \varepsilon)$ by emitting a boson (a plasmon or a particle–hole pair) of momentum $-\boldsymbol{q}$ and energy $\omega = \varepsilon - \xi_m$, and the rate is $2|\operatorname{Im}\Sigma|$. A hole added at $\varepsilon < 0$ decays in the mirror way into an occupied state with $\varepsilon < \xi_m < 0$. An electron at the Fermi level cannot decay, $\operatorname{Im}\Sigma(\varepsilon = 0) = 0$, because the phase space of both processes closes. The real part of Σ follows from the imaginary part by a Kramers–Kronig relation and describes the energy shift of the quasiparticle by virtual emission and reabsorption.

Figure 7 shows how this works numerically for the conduction band of doped graphene at $|\boldsymbol{k}| = 1.5 k_F$. The spectral density of the sample sheet (panel a) has two structures: the sharp optical plasmon of the junction at small q and the broad interband continuum at $\omega > \hbar v_F q$. For

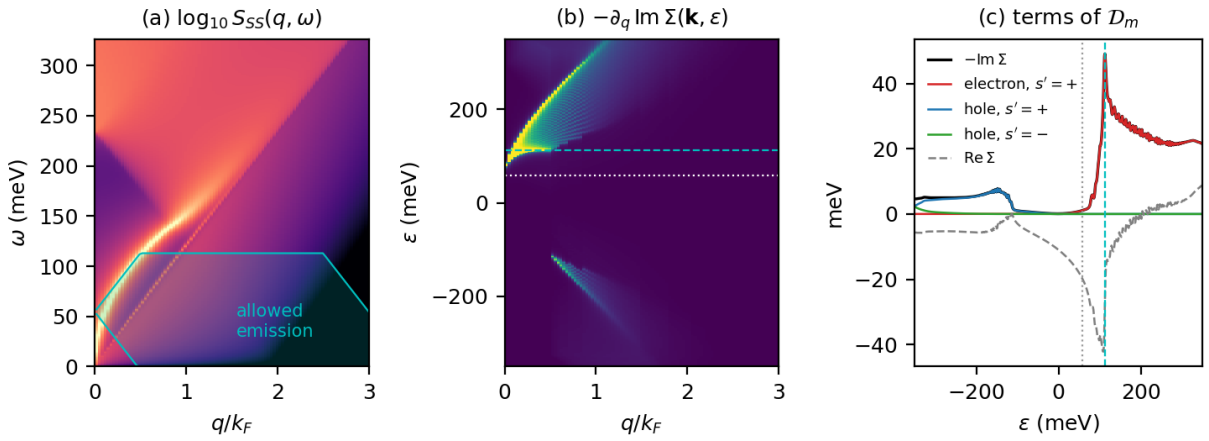


Figure 7: Anatomy of eq. (43) for the conduction band of the SLG sample at $|\boldsymbol{k}| = 1.5k_F$ (SLG | 1 nm hBN | SLG, both sheets at 10^{12} cm^{-2}, $\eta = \eta_\Sigma = 1$ meV). (a) Spectral density $S_{SS}(q,\omega)$ of the sample sheet; the cyan region is the set of (q,ω) that an electron at $\varepsilon = 113$ meV, the peak of $-\operatorname{Im}\Sigma$ (dashed in b and c), can reach by emitting a boson into an empty conduction state. (b) Contribution of each momentum transfer to $-\operatorname{Im}\Sigma$, per unit q/k_F (dotted: bare energy $\xi_{\boldsymbol{k}} = 58$ meV). (c) $-\operatorname{Im}\Sigma$ split into the terms of $\mathcal{D}_m$, labelled by the intermediate band s', and $\operatorname{Re}\Sigma$. The decomposition reproduces the library routine to 10^{-16} (`scripts/plot_notes_selfenergy.py`).

a given ε, the electron term of eq. (47) samples S on the region of the (q,ω) plane allowed by energy conservation and Pauli blocking (cyan). Where this region first meets the plasmon branch, $-\operatorname{Im}\Sigma$ rises steeply: this is the emission edge analysed in section 9.3. Panel (b) resolves $-\operatorname{Im}\Sigma$ by momentum transfer and shows that the edge is built by $q \lesssim 0.6k_F$, while larger momenta contribute only a broad background. Panel (c) separates the terms of $\mathcal{D}_m$. The electron term (m in the conduction band, $\xi_m > 0$) gives the whole of $-\operatorname{Im}\Sigma$ at $\varepsilon > 0$. Below the Fermi level the dominant term is the hole term of the partially filled conduction band ($-E_F < \xi_m < 0$): a hole at $\varepsilon \lesssim -100$ meV can relax inside the Fermi sea by emitting a plasmon, while closer to $\varepsilon = 0$ only bosons of small energy, and hence small weight, are available. The completely filled valence band ($s' = -$) contributes only at the lower end of the window.

5.9 Momentum and frequency quadrature

Equation (43) is evaluated as a finite double sum,

$$\Sigma_{nn'}(\boldsymbol{k},\varepsilon) \simeq -\sum_{\boldsymbol{q}} w_{\boldsymbol{q}} \sum_m \sum_{ab} \rho^a_{nm}(\boldsymbol{k},\boldsymbol{q}) \left[\rho^b_{n'm}(\boldsymbol{k},\boldsymbol{q})\right]^* \sum_{j=1}^{n_\omega} t_j\, S_{ab}(\boldsymbol{q},\omega_j)\, \mathcal{D}_m(\omega_j,\varepsilon), \tag{48}$$

where ω_j are the points of the frequency grid and t_j their trapezoidal weights, $t_j = \frac{1}{2}(\omega_{j+1} - \omega_{j-1})$ with half intervals at the two ends (`compat.trapezoid_weights`). The momentum weights $w_{\boldsymbol{q}}$ (in nm^{-2}) are defined such that $\sum_{\boldsymbol{q}} w_{\boldsymbol{q}} f(\boldsymbol{q})$ approximates $\int \mathrm{d}^2q\, f(\boldsymbol{q})/(2\pi)^2$. Their construction depends on the sample (fig. 8).

Moiré mesh (TBG). A finite periodic sample with N_k moiré unit cells of area A_{cell} has $A = N_k A_{\text{cell}}$ and N_k allowed momenta in the zone, and the area of the zone is $A_{\text{BZ}} = (2\pi)^2/A_{\text{cell}} = \frac{\sqrt{3}}{2}|\boldsymbol{G}_M|^2$. The exact sum of eq. (32) is therefore

$$\frac{1}{A}\sum_{\boldsymbol{q}} f(\boldsymbol{q}) = \frac{A_{\text{BZ}}}{(2\pi)^2}\frac{1}{N_k}\sum_{\boldsymbol{q}} f(\boldsymbol{q}) \xrightarrow{N_k\to\infty} \int_{\text{BZ}} \frac{\mathrm{d}^2q}{(2\pi)^2} f(\boldsymbol{q}), \qquad w_{\boldsymbol{q}} = \frac{A_{\text{BZ}}}{(2\pi)^2}\,\text{weight}(\boldsymbol{q}) \tag{49}$$

(`q_weights_from_band_structure`). Here weight($\boldsymbol{q}$) is the fraction of the zone attributed to the mesh point by `pybandstructure` (`q_sample.weights`, normalised to one). For the $N_s \times N_s$ Wigner–Seitz mesh used here $N_k = N_s^2$ and every point has weight $= 1/N_s^2$, so $w_{\boldsymbol{q}} = A_{\text{BZ}}/(2\pi N_s)^2$ and

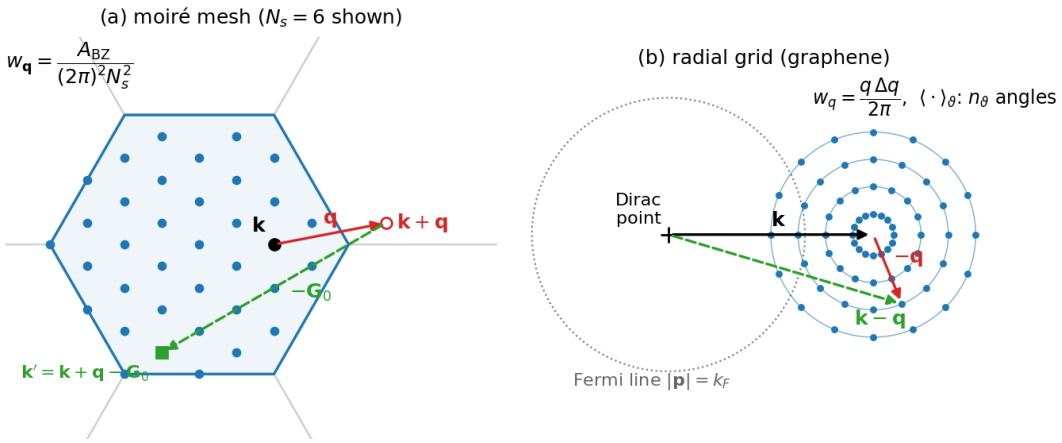


Figure 8: Momentum quadratures. (a) TBG: $\boldsymbol{k}$ and $\boldsymbol{q}$ run over the same $N_s \times N_s$ mesh of the moiré Brillouin zone, every point with the same weight. When $\boldsymbol{k}+\boldsymbol{q}$ falls outside the zone it is folded back by a moiré vector $\boldsymbol{G}_0$, which the form factor (27) compensates by shifting the plane-wave index. (b) Graphene: the spectral density depends only on $q = |\boldsymbol{q}|$, so $\boldsymbol{q}$ is sampled on rings of radius q_i around the external momentum, with n_ϑ directions per ring, and the intermediate states sit at $\boldsymbol{k}-\boldsymbol{q}$.

$\sum_{\boldsymbol{q}} w_{\boldsymbol{q}} = 1/A_{\text{cell}}$. The weight contains no spin or valley degeneracy (section 5.11). The external momentum $\boldsymbol{k}$ must itself be a mesh point, so that $\boldsymbol{k}+\boldsymbol{q}$ folds onto the mesh. The point $\boldsymbol{q}=0$ is removed: there V diverges, but the singularity is integrable ($\int \mathrm{d}^2q/q \propto \int \mathrm{d}q$), and dropping the cell around $\boldsymbol{q}=0$ changes Σ by an amount proportional to its linear size $|\boldsymbol{G}_M|/N_s$, which vanishes as the mesh is refined.

Radial grid (isotropic samples). For graphene $S(\boldsymbol{q},\omega)$ depends only on $|\boldsymbol{q}|$, while the intermediate energy $\xi^{s'}_{\boldsymbol{k}-\boldsymbol{q}} = s'\hbar v_F|\boldsymbol{k}-\boldsymbol{q}| - E_F$ and the chirality factor depend on the direction of $\boldsymbol{q}$ relative to $\boldsymbol{k}$. The measure is split into a radial and an angular part,

$$\int \frac{\mathrm{d}^2q}{(2\pi)^2} f(\boldsymbol{k},\boldsymbol{q}) = \int_0^{q_{\max}} \frac{q\,\mathrm{d}q}{2\pi} \left\langle f(\boldsymbol{k},\boldsymbol{q}(\vartheta))\right\rangle_\vartheta, \qquad \langle f\rangle_\vartheta = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\vartheta\, f \simeq \frac{1}{n_\vartheta}\sum_{j=0}^{n_\vartheta-1} f(\vartheta_j), \tag{50}$$

with $\vartheta_j = 2\pi j/n_\vartheta$ (`make_angular_quadrature`) and the radial weights $w_{q_i} = q_i\,\Delta q_i/2\pi$, $\Delta q_i = \frac{1}{2}(q_{i+1} - q_{i-1})$ and half intervals at the ends (`make_radial_q_weights`). The code places the intermediate state at $\boldsymbol{k}-\boldsymbol{q}$ instead of $\boldsymbol{k}+\boldsymbol{q}$, which is equivalent because $\boldsymbol{q}\to-\boldsymbol{q}$ leaves both the measure and $S(|\boldsymbol{q}|,\omega)$ unchanged. The angular average cannot be skipped: sampling only $\vartheta = 0$ (collinear $\boldsymbol{q}$) computes a different quantity, not a coarse version of the same one. Defaults: $q \in [0.02, 3]k_F$ with $n_q = 96$, $n_\vartheta = 64$, and $\omega \in (0, 2.8E_F]$ with $n_\omega = 480$.

5.10 Causality

The anti-Hermitian part of eq. (43) is, for $\eta_\Sigma \to 0$,

$$\frac{\Sigma - \Sigma^\dagger}{2\mathrm{i}} = -\pi \sum_{\boldsymbol{q}} w_{\boldsymbol{q}} \sum_m \Big[N^+_m\, \mathcal{F}_m\, S(\boldsymbol{q}, \varepsilon - \xi_m)\, \mathcal{F}^\dagger_m + N^-_m\, \mathcal{F}_m\, S(\boldsymbol{q}, \xi_m - \varepsilon)\, \mathcal{F}^\dagger_m \Big], \tag{51}$$

with the vertex matrix $(\mathcal{F}_m)_{na} = \rho^a_{nm}(\boldsymbol{k},\boldsymbol{q})$, the populations evaluated at the frequency where S is taken, and $S(\boldsymbol{q},\omega)$ understood to vanish for $\omega < 0$. It is a sum of negative semidefinite matrices when $S \succeq 0$ and $N^\pm \geq 0$. Hence $\mathrm{Im}\,\Sigma_{nn} \leq 0$ and the spectral function of section 6 is positive. This holds exactly with the Hermitian spectral density of eq. (34) and is verified numerically (`spectral.is_passive`).

5.11 Spin and valley degeneracy

The Coulomb potential varies on the scale of the interlayer distance, much longer than the lattice constant, so a scattering event conserves spin and does not transfer an electron between valleys, which would require a momentum $|\boldsymbol{K} - \boldsymbol{K}'| \sim 1/a$ at which V is negligible [19]. The polarisation bubble (15) is a closed fermion loop and sums over the $g = 4$ flavours of the particle–hole pair, so g enters χ^0, ε and W. In eq. (32), instead, the internal electron line is the continuation of the external one and carries its flavour. There is no free flavour sum and hence *no* degeneracy factor in $w_{\boldsymbol{q}}$. Including one would make Σ four times too large. Spin and valley enter the self-energy only through the screening.

5.12 What is not included: exchange and double counting

Only the correlation part is computed. The spectral density S is that of W, and V has no spectral weight for a static background, so the exchange term (30) is absent. Σ^x is real and static. It shifts and reshapes the quasiparticle bands but cannot produce satellites or finite lifetimes. Ref. [1] (Sec. IV B) likewise neglects the first-order Hartree and Fock diagrams, with a normal-ordered Coulomb interaction. The consequences differ between the two samples.

For TBG at neutrality the Hartree potential of the BM model vanishes, while the Fock term would renormalise the flat bands. The correlation-only Σ is therefore the self-energy of the scheme of Ref. [1], and the quasiparticle peaks are the bare continuum bands dressed by $\mathrm{Re}\,\Sigma_c$ only, which should be kept in mind when comparing with experiment.

For doped graphene the omitted Σ^x is what enhances the Fermi velocity [19, 31]. Without it the correlation part alone narrows the band ($v^*/v_F \simeq 0.92$ in notebook 02). Since Σ^x is real, it leaves $\mathrm{Im}\,\Sigma$, and hence the energy of the satellite structure, unchanged. It does move the quasiparticle peak by $\Sigma^x_{nn}(\boldsymbol{k})$ and therefore changes the offset between peak and satellite. Quasiparticle velocities and peak–satellite offsets of doped graphene computed here are correlation-only quantities. Σ^x can be added to the Dyson equation as a static term when needed.

5.13 Plasmon-pole form

Inserting $S(\boldsymbol{q},\omega) \to \sum_p R_p(\boldsymbol{q})\,\delta(\omega - \omega_p(\boldsymbol{q}))$ (section 4.4) in eq. (43) gives

$$\Sigma^{\mathrm{PPA}}_{nn'}(\boldsymbol{k},\varepsilon) = -\sum_{\boldsymbol{q}} w_{\boldsymbol{q}} \sum_{m,p} \sum_{ab} \rho^a_{nm} \left[\rho^b_{n'm}\right]^* R_{p,ab}(\boldsymbol{q})\, \mathcal{D}_m(\omega_p(\boldsymbol{q}),\varepsilon) \tag{52}$$

(`compute_self_energy_layer`, `compute_self_energy_pybandstructure`), where p labels the collective modes at $\boldsymbol{q}$, $\omega_p(\boldsymbol{q})$ their energies and $R_p(\boldsymbol{q})$ their residue matrices, eq. (20). The full residue matrix is used. A rank-one factorisation $R_p \simeq \Phi_p \Phi_p^\dagger$ is available as a diagnostic only. The plasmon-pole approximation [15, 16] ignores the particle–hole continuum, which carries most of the weight of S in the regimes studied here (section 4.4).

5.14 Fast route: tabulated resolvent kernels

In eqs. (43) and (44) the energies ξ_m and ε appear in the denominators only through $u = \xi_m - \varepsilon$, while the populations split as $N_m^{\pm} = n_{\mathrm{B}}(\omega) + n_{\mathrm{F}}(\mp\xi_m)$. Therefore

$$\int_0^{\omega_{\max}} \mathrm{d}\omega\, S(\boldsymbol{q},\omega)\mathcal{D}_m(\omega,\varepsilon) = n_{\mathrm{F}}(-\xi_m)\, P_{\boldsymbol{q}}(u) + n_{\mathrm{F}}(\xi_m)\, M_{\boldsymbol{q}}(u) + B^+_{\boldsymbol{q}}(u) + B^-_{\boldsymbol{q}}(u), \tag{53}$$

$$P_{\boldsymbol{q}}(u) = \int \mathrm{d}\omega \frac{S(\boldsymbol{q},\omega)}{\omega + u - \mathrm{i}\eta_\Sigma}, \quad M_{\boldsymbol{q}}(u) = \int \mathrm{d}\omega \frac{S(\boldsymbol{q},\omega)}{-\omega + u - \mathrm{i}\eta_\Sigma}, \quad B^{\pm}_{\boldsymbol{q}}(u) = \int \mathrm{d}\omega \frac{n_{\mathrm{B}}(\omega) S(\boldsymbol{q},\omega)}{\pm\omega + u - \mathrm{i}\eta_\Sigma}. \tag{54}$$

These four matrix-valued functions of a single variable depend on $\boldsymbol{q}$ but not on $\boldsymbol{k}$, m or ε. They are tabulated once on a uniform u grid of spacing $\eta_\Sigma/8$, with the same trapezoidal rule in ω, and linearly interpolated for every $(\boldsymbol{k}, m, \varepsilon)$ (`ResolventKernel`). This is an exact rewriting up to the

interpolation error ($\lesssim 2\times10^{-4}$ relative in the tests) and makes the cost per external $\boldsymbol{k}$ independent of n_ω. For graphene it is about 300 times faster than the direct contraction, which makes full twist maps cheap. At $T=0$ the Bose terms vanish because the ω grid starts above zero.

5.15 Numerical parameters and their roles

symbol	code	role
η	`cfg.eta`, `TBGParams.eta`	broadening of χ^0 and V ($\omega+\mathrm{i}\eta$), sets the width of undamped plasmons in S. TBG default: $2\hbar v_{\mathrm{BM}}\lvert\boldsymbol{G}_M\rvert/N_s$, with v_{BM} the flat-band velocity estimate of eq. (13) (valid only in the flat-band regime)
η_Σ	`eta_self`	finite value of the 0^+ in eq. (44), floors $\mathrm{Im}\,\Sigma$ near $\varepsilon=0$ (the Fermi-liquid ε^2 law is recovered only for $\lvert\varepsilon\rvert\gg\eta_\Sigma$)
η_{qp}	`eta_qp`	extrinsic width in the Dyson equation (finite junction, disorder), keeps the quasiparticle peak finite
$\omega_{\max}$, n_ω	`omega_grid`	frequency window of S, must contain the plasmon branches and the relevant continuum
$q_{\max}$, n_q, n_ϑ	radial grid	graphene: $3k_F$, 96, 64 (convergence is slowest in n_q, error $\sim 1/n_q$)
N_s, n_b	`n_sites`, `n_bands`	TBG mesh and number of bands per side of neutrality; memory of `densityL` $\propto 2(2n_b)^2N_s^4$

6 Dyson equation and spectral function

6.1 Green's function and spectral function

With the bare propagator $G^0(\boldsymbol{k},\varepsilon)=\sum_n|n\boldsymbol{k}\rangle\langle n\boldsymbol{k}|/(\varepsilon+\mathrm{i}\eta_{\mathrm{qp}}-\xi_{n\boldsymbol{k}})$ and the self-energy matrix of eq. (43), the Dyson equation $G=G^0+G^0\Sigma G$ restricted to a set of n_b bands reads (Ref. [1], Eq. (50))

$$G(\boldsymbol{k},\varepsilon)=\left[(\varepsilon+\mathrm{i}\eta_{\mathrm{qp}})\mathbb{1}-\mathrm{diag}(\xi_{n\boldsymbol{k}})-\Sigma(\boldsymbol{k},\varepsilon)\right]^{-1},\qquad A(\boldsymbol{k},\varepsilon)=\frac{\mathrm{i}}{2\pi}\left[G(\boldsymbol{k},\varepsilon)-G(\boldsymbol{k},\varepsilon)^\dagger\right] \tag{55}$$

(`spectral.green_function_matrix`, `spectral_function_matrix`). A is Hermitian and positive semidefinite whenever $(\Sigma-\Sigma^\dagger)/2\mathrm{i}\preceq 0$ (section 5.10), because $A=\frac{1}{\pi}G\,\Gamma\,G^\dagger$ with $\Gamma=\eta_{\mathrm{qp}}\mathbb{1}-(\Sigma-\Sigma^\dagger)/2\mathrm{i}\succeq 0$. Its trace obeys $\int\mathrm{d}\varepsilon\,\mathrm{Tr}\,A=n_b$ (checked to 10^{-3} for $\Sigma=0$, `test_spectral_function_normalised_without_interactions`). If Σ is diagonal (band-diagonal approximation, or exactly for isotropic graphene, where the chirality structure makes Σ diagonal in the band basis) this reduces to

$$A_n(\boldsymbol{k},\varepsilon)=-\frac{1}{\pi}\,\mathrm{Im}\,\frac{1}{\varepsilon-\xi_{n\boldsymbol{k}}-\Sigma_{nn}(\boldsymbol{k},\varepsilon)+\mathrm{i}\eta_{\mathrm{qp}}}=\frac{1}{\pi}\,\frac{\eta_{\mathrm{qp}}-\mathrm{Im}\,\Sigma_{nn}}{(\varepsilon-\xi_{n\boldsymbol{k}}-\mathrm{Re}\,\Sigma_{nn})^2+(\eta_{\mathrm{qp}}-\mathrm{Im}\,\Sigma_{nn})^2} \tag{56}$$

(`spectral.spectral_function_scalar`, `slg_self_energy.spectral_function_sample`). The band-off-diagonal elements matter where two bands are close compared with $|\Sigma|$, e.g. inside the flat-band manifold of TBG, where $\|\Sigma_{\mathrm{off}}\|/\|\Sigma_{\mathrm{diag}}\|\simeq 0.3$ at κ_{top} (notebook 03). Within an exactly degenerate pair of bands the split into diagonal and off-diagonal parts depends on the gauge, and only the matrix (or a gauge-invariant contraction such as eq. (62)) is meaningful.

6.2 Reading the spectral function

Quasiparticle peak. Near the solution $E^*_{n\boldsymbol{k}}$ of

$$E^*-\xi_{n\boldsymbol{k}}-\mathrm{Re}\,\Sigma_{nn}(\boldsymbol{k},E^*)=0 \tag{57}$$

A is a Lorentzian of weight and width

$$Z=\left[1-\partial_\varepsilon\,\mathrm{Re}\,\Sigma_{nn}\right]^{-1}_{E^*},\qquad \Gamma=2Z\left(\eta_{\mathrm{qp}}-\mathrm{Im}\,\Sigma_{nn}(E^*)\right) \tag{58}$$

(`spectral.quasiparticle_solution`). The slope of $E^*_{n\boldsymbol{k}}$ as a function of $|\boldsymbol{k}|$ defines the renormalised (quasiparticle) velocity v^* quoted for graphene. The peak does not sit at the bare band: $\mathrm{Re}\,\Sigma$ displaces it, and the missing weight $1-Z$ is transferred to the incoherent part and to the satellites.

Satellites. Away from the quasiparticle pole, to first order in Σ (Ref. [1], Eq. (34)),

$$A_{\lambda_1\lambda_2}(\boldsymbol{k},\varepsilon) \simeq -\frac{1}{\pi}\frac{\Sigma''_{\lambda_1\lambda_2}(\boldsymbol{k},\varepsilon)}{(\varepsilon-\xi_{\lambda_1\boldsymbol{k}})(\varepsilon-\xi_{\lambda_2\boldsymbol{k}})}, \qquad \Sigma'' = \frac{\Sigma-\Sigma^\dagger}{2\mathrm{i}}, \tag{59}$$

so peaks of $-\mathrm{Im}\,\Sigma$ produce satellites [14, 17]. In doped graphene they are the plasmarons predicted in RPA-G_0W [30, 21] and observed in photoemission [20]. From eq. (43), $-\mathrm{Im}\,\Sigma$ is large where an electron at ε can emit a boson: $\varepsilon = \xi_{m\boldsymbol{k}+\boldsymbol{q}} + \omega$ with $S(\boldsymbol{q},\omega)$ large. A dispersionless mode of energy ω_0 (the flat-band plasmon of TBG) therefore gives a satellite at $\varepsilon \simeq \xi_{\mathrm{VH}} + \omega_0$ where the intermediate density of states peaks (ξ_{VH} the van Hove energy of the flat band), nearly independent of $\boldsymbol{k}$. For a dispersive mode (the $\sqrt{q}$ plasmon of graphene) the satellite sits at $\varepsilon = \xi_{\boldsymbol{k}-\boldsymbol{q}} + \omega_{\boldsymbol{q}}$ summed over $\boldsymbol{q}$. Since $\xi_{\boldsymbol{k}-\boldsymbol{q}} \neq \xi_{\boldsymbol{k}}$, its offset from the quasiparticle peak is not the plasmon energy at any single $\boldsymbol{q}$. How to estimate it from the plasmon dispersion is shown in section 9.3. For one-boson emission the satellite lies on the side of the quasiparticle peak away from E_F (above it for empty states, below it for occupied ones).

Fermi-liquid region. At $\varepsilon \to 0$, $-\mathrm{Im}\,\Sigma \propto \varepsilon^2$ (section 5.8). On a numerical grid this is floored by η_Σ and by the finite q mesh, so the region $|\varepsilon| \lesssim 20\eta_\Sigma$ is not physical.

7 Quantum twisting microscope: differential conductance

In the QTM [3] a graphene tip is pressed onto the sample through a thin hBN barrier and rotated by an angle θ, and the tunneling current is recorded as a function of bias V and θ. Because tunneling across a clean, flat interface conserves in-plane crystal momentum, the twist selects *which momentum of the sample* is probed, and the bias selects the energy: the (θ, V) map is a momentum- and energy-resolved picture of the sample spectral function [2, 1]. This section derives the expression used by `g0w.qtm` and describes the kinematics for the SLG and TBG samples.

7.1 From the tunneling Hamiltonian to dI/dV

Tunneling Hamiltonian. Let $c_{\boldsymbol{k}\sigma}$ annihilate a tip electron of lab momentum $\boldsymbol{k}$ on sublattice σ and $d_{\boldsymbol{p}\sigma'}$ a top-layer electron of the sample. For a junction of area A_{J} much larger than the lattice constants, the tunneling amplitude conserves lab momentum modulo the reciprocal lattices of the two crystals, $\boldsymbol{p}+\boldsymbol{G}_S = \boldsymbol{k}+\boldsymbol{G}_T$ with $\boldsymbol{G}_T = O(\theta)\boldsymbol{G}_S$ (Ref. [1], Eq. (30)), where $\boldsymbol{G}_S$ and $\boldsymbol{G}_T$ are reciprocal-lattice vectors of the sample and of the tip and $O(\theta)$ is the clockwise rotation $\left(\begin{smallmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{smallmatrix}\right)$ (`qtm.rotation_matrix`). Its Fourier components fall off on the atomic scale, so only the processes with $|\boldsymbol{k}+\boldsymbol{G}_T| \simeq |\boldsymbol{p}+\boldsymbol{G}_S| \simeq K = 4\pi/3a$ survive: three Bragg channels $j=1,2,3$ linking the three equivalent corners of the two Brillouin zones. In channel j the sublattice structure is $t^{(j)}_{\sigma\sigma'} = w\, e^{\mathrm{i}(\boldsymbol{G}^j_T\cdot\boldsymbol{\tau}_\sigma - \boldsymbol{G}^j_S\cdot\boldsymbol{\tau}_{\sigma'})}$, where w is the tunneling amplitude of one Bragg channel and $\boldsymbol{\tau}_\sigma$ the position of sublattice σ in the unit cell. With the origin on an atom the representative channel $j=1$ has all phases equal to one,

$$t^{(1)} = w\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} = w\,|\chi\rangle\langle\chi|, \qquad |\chi\rangle = (1,1)^T. \tag{60}$$

For a C_3-symmetric sample the three channels probe C_3-related momenta with C_3-related matrix elements and contribute equally (factor $N_b = 3$). The tip couples only to the layer it touches (the top layer of TBG): tunneling decays exponentially with distance.

Current. To lowest order in w (assumption 5) the current from tip to sample is the golden-rule expression

$$I = \frac{2\pi e}{\hbar} N_f N_b \sum_{\boldsymbol{k}} \int \mathrm{d}\varepsilon \, [n_\mathrm{F}(\varepsilon - \mu_T) - n_\mathrm{F}(\varepsilon - \mu_S)] \; \mathrm{Tr}[t^{(1)\dagger} A_T(\boldsymbol{k}, \varepsilon) \, t^{(1)} A_S^{\mathrm{top}}(\boldsymbol{p}(\boldsymbol{k}), \varepsilon)], \tag{61}$$

with $N_f = 4$ spin–valley flavours, μ_T and μ_S the chemical potentials of tip and sample, A_T the 2×2 (sublattice) spectral function of the tip and A_S^{top} the top-layer sublattice block of the sample spectral function at the conserved momentum $\boldsymbol{p}(\boldsymbol{k})$. Using eq. (60), the trace factorises into $w^2 \langle\chi|A_T|\chi\rangle \, \langle\chi|A_S^{\mathrm{top}}|\chi\rangle$.

Bias and the tip as a momentum filter. Following Ref. [1], (a) the tip is non-interacting graphene, and its chemical potential follows the bias while its band energies stay fixed relative to the sample Fermi level, $\partial(\xi^T + eV)/\partial V \simeq 0$: the bias changes only the occupations in eq. (61), $\mu_T = \mu_S + eV$. (b) The sample chemical potential does not move relative to the sample bands (for TBG because of the large flat-band density of states, for the doped SLG sample because it is fixed by its own gate). (c) $T = 0$. Differentiating eq. (61) with respect to V then gives $dI/dV \propto \sum_{\boldsymbol{k}} \mathrm{Tr}[t^{(1)\dagger} A_T(\boldsymbol{k}, \mu_T) \, t^{(1)} A_S^{\mathrm{top}}(\boldsymbol{p}(\boldsymbol{k}), \mu_S + eV)]$: only tip states *at the tip Fermi level* contribute, and they lie on the tip Fermi line around the tip Dirac point $\boldsymbol{K}_\theta = O(\theta)\boldsymbol{K}$. (d) The tip Fermi momentum is small, so that A_S is constant along this line and is evaluated at $\boldsymbol{p} = \boldsymbol{K}_\theta$. What remains of the tip is $\sum_{\boldsymbol{k}} \langle\chi|A_T(\boldsymbol{k}, \mu_T)|\chi\rangle = A_\mathrm{J} \, \nu_T \, \langle|\langle\chi|u_T\rangle|^2\rangle = A_\mathrm{J} \, \nu_T$: the average of $|1 + s e^{\mathrm{i}\phi}|^2/2$ over the tip Fermi line equals one, so the tip spinor disappears and only the tip density of states ν_T (per unit area and per flavour, at the tip Fermi level) survives. The three Bragg channels do not interfere at generic (incommensurate) angles and add up to the factor N_b. The result is (Ref. [1], Eqs. (8) and (33))

$$\boxed{\frac{dI}{dV}(V, \theta) = e^2 \Gamma \sum_{\lambda\rho} \Lambda_\lambda(\boldsymbol{K}_\theta) \, A^S_{\lambda\rho}(\boldsymbol{K}_\theta, eV) \, \Lambda_\rho(\boldsymbol{K}_\theta)^* = e^2 \Gamma \, \langle\chi_\theta|A^S(eV)|\chi_\theta\rangle}, \quad \Gamma = \tfrac{2\pi}{\hbar} w^2 A_\mathrm{J} \, \nu_T N_b N_f \tag{62}$$

(Γ of Ref. [1], Eq. (8), times N_f), where eV is measured from μ_S and is the energy axis of A^S (bias and sample energy are the same axis: `eps_values` $= eV$). The band-basis spectral function of eq. (55) enters through $A_S^{\mathrm{top}} = \sum_\lambda |u_\lambda^{\mathrm{top}}\rangle A_{\lambda\rho} \langle u_\rho^{\mathrm{top}}|$, so the tunneling amplitude is the sublattice sum of the top-layer spinor of the Bloch state (Ref. [1], Eq. (32)),

$$\Lambda_\lambda(\boldsymbol{K}_\theta) = \langle\chi|u_\lambda^{\mathrm{top}}(\boldsymbol{K}_\theta)\rangle = u^\lambda_{\boldsymbol{K}_\theta, tA} + u^\lambda_{\boldsymbol{K}_\theta, tB}. \tag{63}$$

The code returns $dI/dV/(e^2\Gamma)$ in eV^{-1} and drops Γ.

Properties. (i) Positivity: $dI/dV = e^2\Gamma\langle\chi_\theta|A^S|\chi_\theta\rangle \geq 0$ because A^S is PSD (section 6). (ii) Gauge invariance: under $u_\lambda \to e^{\mathrm{i}\varphi_\lambda} u_\lambda$, $\Lambda_\lambda \to e^{\mathrm{i}\varphi_\lambda}\Lambda_\lambda$ and $A_{\lambda\rho} \to e^{-\mathrm{i}\varphi_\lambda} A_{\lambda\rho} e^{\mathrm{i}\varphi_\rho}$, so eq. (62) is invariant *provided* Λ *and* Σ *are built from the same eigenvectors* (`test_band_matrix_didv_is_gauge_invariant_and_positive`). With the band-diagonal approximation the off-diagonal terms are dropped and only $|\Lambda_\lambda|^2$ enters, which is gauge invariant by itself. (iii) Non-interacting limit: $dI/dV = \sum_\lambda |\Lambda_\lambda|^2 \delta_\eta(eV - \xi_{\lambda\boldsymbol{K}_\theta})$ (`tbg_bare_didv`).

What eq. (62) leaves out. The radius of the tip Fermi line, $k_{F,T}$, sets the momentum resolution, which eq. (62) treats as perfect. If the tip bands do move with the bias (a tip with fixed doping, whose Dirac point follows μ_T), the derivative of eq. (61) acquires a second term from tip states below the tip Fermi level, on rings $|\boldsymbol{k} - \boldsymbol{K}_\theta| = |\varepsilon - E_{D,T}|/\hbar v$, where $E_{D,T}$ is the energy of the tip Dirac point. This term is dropped by assumption (a). Self-energy effects of the tip, inelastic (boson-assisted) tunneling as a separate channel, and the bias dependence of the electrostatics are not included: interactions enter only through the dressed sample spectral function (assumption 5), whose plasmon satellites are, in this language, the inelastic features. At finite temperature eq. (62) is convolved with $-\partial_\varepsilon n_\mathrm{F}$, a bell of standard deviation $\pi k_B T/\sqrt{3} \simeq 1.8 k_B T$ (0.8 meV at 5 K). `qtm.smooth_bias` applies a Gaussian of chosen width, which also mimics the lock-in modulation.

7.2 SLG sample

Tip and sample are both graphene with Dirac point $\boldsymbol{K} = (K, 0)$ at $\theta = 0$. Measured from the sample Dirac point the probed momentum is (`qtm.slg_probe_momentum`)

$$\boldsymbol{p}_\theta = \boldsymbol{K}_\theta - \boldsymbol{K} = (O(\theta) - 1)\boldsymbol{K}, \qquad |\boldsymbol{p}_\theta| = 2K \sin\tfrac{\theta}{2} \simeq K\theta = 0.297 \text{ nm}^{-1} \text{ per degree}, \tag{64}$$

almost perpendicular to $\boldsymbol{K}$. One degree already corresponds to $1.7k_F$ at $n = 10^{12}$ cm^{-2}. The isotropic cone makes Σ diagonal in the chirality basis (section 6), and with the spinor $(1, se^{\mathrm{i}\phi_{\boldsymbol{p}}})/\sqrt{2}$

$$\frac{dI}{dV} = e^2\Gamma \sum_{s=\pm} |\Lambda_s(\boldsymbol{p}_\theta)|^2 A_s^S(\boldsymbol{p}_\theta, eV), \qquad |\Lambda_s|^2 = 1 + s\cos\phi_{\boldsymbol{p}_\theta} \tag{65}$$

(`qtm.slg_spectral_map`, `slg_didv`). Since $\phi_{\boldsymbol{p}_\theta} \simeq -\frac{\pi}{2} - \frac{\theta}{2}$, $\cos\phi_{\boldsymbol{p}_\theta} = -\sin\frac{\theta}{2}$ and the two bands are probed with nearly equal weight (difference $< 4\%$ for $\theta < 2°$). The option `tunneling_model="constant"` sets $|\Lambda_s|^2 = 1$ and reads A^S directly. The map shows two quasiparticle ridges $eV = \pm\hbar v^*|\boldsymbol{p}_\theta| - E_F$ (renormalised velocity v^*, section 6) and the plasmaron satellites displaced from them. The twist sets the *external* momentum $\boldsymbol{p}_\theta$, while the plasmon momentum is integrated over inside $\Sigma(\boldsymbol{p}_\theta, \varepsilon)$ and must not be confused with $\boldsymbol{p}_\theta$. The angle can be converted to momentum and back with `qtm.twist_to_momentum` and `momentum_to_twist`.

7.3 TBG sample

Kinematics. Now θ is the twist between the tip and the *top* layer of TBG, and the tip couples to the top-layer plane-wave components. A top-layer plane wave $\boldsymbol{k} + \boldsymbol{G}$ of the continuum model carries the lab momentum $\boldsymbol{k} + \boldsymbol{G} + \boldsymbol{\Gamma}^{\text{lab}}$ (eq. (14)). Momentum conservation with the tip Dirac point $\boldsymbol{K}_\theta^{\text{lab}} = O(\theta)\boldsymbol{K}_{\text{top}}^{\text{lab}}$ then fixes the extended-zone model momentum

$$\boxed{\boldsymbol{k}_\theta = O(\theta)\,\boldsymbol{K}_{\text{top}}^{\text{lab}} - \boldsymbol{\Gamma}^{\text{lab}}} \qquad (\texttt{qtm.tbg_tip_momentum}), \tag{66}$$

which passes through

$$\theta = 0:\ \boldsymbol{k}_\theta = \boldsymbol{K}_1\ (\kappa_{\text{top}}), \qquad \theta = \theta_{\text{TBG}}:\ \boldsymbol{k}_\theta = \boldsymbol{K}_2\ (\kappa_{\text{bot}}), \qquad \theta = -\theta_{\text{TBG}}:\ \boldsymbol{k}_\theta \simeq 2\boldsymbol{K}_1 - \boldsymbol{K}_2\ (\gamma'), \tag{67}$$

the first two exactly (both Dirac points lie on the circle $|\boldsymbol{k} + \boldsymbol{\Gamma}^{\text{lab}}| = K$ swept by the tip), the last to $O(\theta^2)$: it is the γ point of a neighbouring moiré zone ($|2\boldsymbol{K}_1 - \boldsymbol{K}_2| = |\boldsymbol{G}_M|$), missed by 0.6% at 1.2°. To $O(\theta^2)$ the trajectory is a straight line through κ_{top} along $\boldsymbol{K}_2 - \boldsymbol{K}_1$ traversed at K per radian, i.e. one side $|\boldsymbol{K}_1 - \boldsymbol{K}_2| = |\boldsymbol{G}_M|/\sqrt{3}$ of the moiré hexagon per θ_{TBG}. Physically: when the tip is aligned with the top layer it probes the top-layer Dirac point, when it is aligned with the bottom layer the bottom one.

Folding and unfolding. The band structure lives on the mesh of the first moiré zone, so $\boldsymbol{k}_\theta$ is decomposed as $\boldsymbol{k}_\theta = \boldsymbol{k} + \boldsymbol{G}$ with $\boldsymbol{k}$ the nearest mesh point and $\boldsymbol{G}$ a moiré vector (`tbg_model.fold_to_mesh`). The snapping error is reported and is at most $1/\sqrt{3}$ of a mesh spacing. The spectral function $A^S_{\lambda\rho}(\boldsymbol{k}, eV)$ is evaluated at the folded $\boldsymbol{k}$, but the tunneling amplitude is the *unfolded* plane-wave component,

$$\Lambda_\lambda(\boldsymbol{k}_\theta) = u^\lambda_{\boldsymbol{k}}[4g + 0] + u^\lambda_{\boldsymbol{k}}[4g + 1], \qquad g = \text{index of } \boldsymbol{G} \tag{68}$$

(`qtm.tbg_tunneling_amplitudes`), and it is zero if $\boldsymbol{k} + \boldsymbol{G}$ lies outside the plane-wave cutoff. This is what makes the QTM an extended-zone probe: the same folded state $\boldsymbol{k}$ is seen with different weights at $\boldsymbol{k}_\theta = \boldsymbol{k} + \boldsymbol{G}$ for different $\boldsymbol{G}$, and a band that is mostly bottom-layer at $\boldsymbol{k}_\theta$ is invisible to the tip. $|\Lambda_\lambda|^2$ is not the top-layer weight: it is the coherent sublattice sum of one plane-wave component, and it varies by orders of magnitude between bands at the same $\boldsymbol{k}$.

Validation. With the interlayer hopping switched off, the top layer is a Dirac cone centred at $\kappa_{\rm top}$ in the extended zone, and the bare conductance must follow $eV = \pm\hbar v\, 2K \sin(\theta/2)$ for every θ, also outside the first moiré zone. The kinematics of eq. (66) with the unfolded amplitude reproduces it (`test_decoupled_layers_follow_the_extended_zone_dirac_cone`, notebook 05, Sec. 2).

Interband terms and gauge. The band-diagonal approximation (`band_diagonal=True`) keeps only $\Sigma_{\lambda\lambda}$, and then only $|\Lambda_\lambda|^2$ enters. It is accurate where bands are separated by more than $|\Sigma|$ and fails inside the flat-band manifold and at band crossings. With the full band matrix the eigenvectors that build Λ must be those that build $\Sigma_{\lambda\rho}$ (the stored `band_structure.wavefunctions`): a separate diagonalisation would assign independent phases to the two and spoil the gauge invariance of eq. (62).

Second derivative. Peaks of A^S appear as peaks of dI/dV. Their edges, and steps from the onset of a decay channel, are sharpened in d^2I/dV^2, which changes sign at every maximum of dI/dV (`qtm.bias_derivative`, after `smooth_bias`). It is the quantity usually shown for weak inelastic features, here the plasmon satellites.

8 Worked examples

The figures of this section are produced by the notebooks (`notebooks/01`–`05`) with the parameters of section 10, and the numbers quoted are printed by the notebooks.

8.1 Electrostatics of the junction (notebook 01)

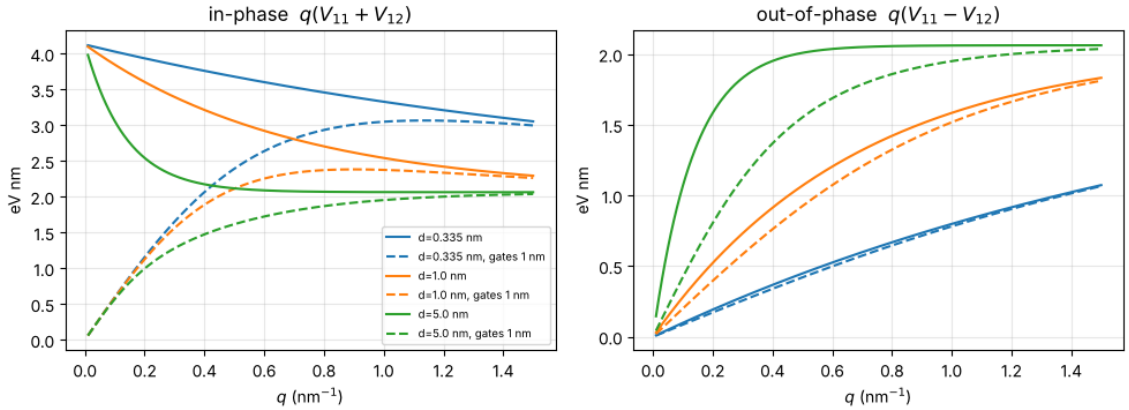


Figure 9: In-phase and out-of-phase Coulomb channels $q(V_{11} \pm V_{12})$ of two sheets in hBN ($\varepsilon_\parallel = 6.41$, $\varepsilon_z = 3$) for three separations, without (solid) and with (dashed) gates 1 nm away. Without gates $qV_+ \to 4\pi e^2/\bar\varepsilon = 4.13$ eV nm at $q \to 0$ and both tend to $2\pi e^2/\bar\varepsilon$ at $q \gg 1/d$. The out-of-phase channel vanishes linearly, $qV_- \simeq 2\pi e^2 qd/\varepsilon_z$. Gates kill the long-range in-phase part and barely touch the out-of-phase one.

The closed forms of section 2 are reproduced to machine precision (e.g. $V_{12}/V_{11} = e^{-\kappa d} = 0.8634$ at $d = 0.335$ nm), the dispersive hBN kernel is passive in the Reststrahlen bands, and the capped-slab kernel interpolates between its limits (fig. 9).

8.2 Doped monolayer graphene (notebook 02)

The SLG|1 nm hBN|SLG junction with $n = 10^{12}$ cm^{-2} in both sheets ($k_F = 0.177$ nm^{-1}, $E_F = 116.6$ meV, $\bar\varepsilon = 4.39$, graphene fine-structure constant $e^2/\bar\varepsilon\hbar v_F = 0.50$). The closed-form χ^0 agrees with a brute-force Lindhard sum to $< 6 \times 10^{-3} D(E_F)$.

The kernel route reproduces the direct frequency integration to 1.3×10^{-4} (1.1 s against 78 s per $\boldsymbol{k}$). The plasmaron satellites of fig. 12 are the features probed by the QTM in section 8.4.

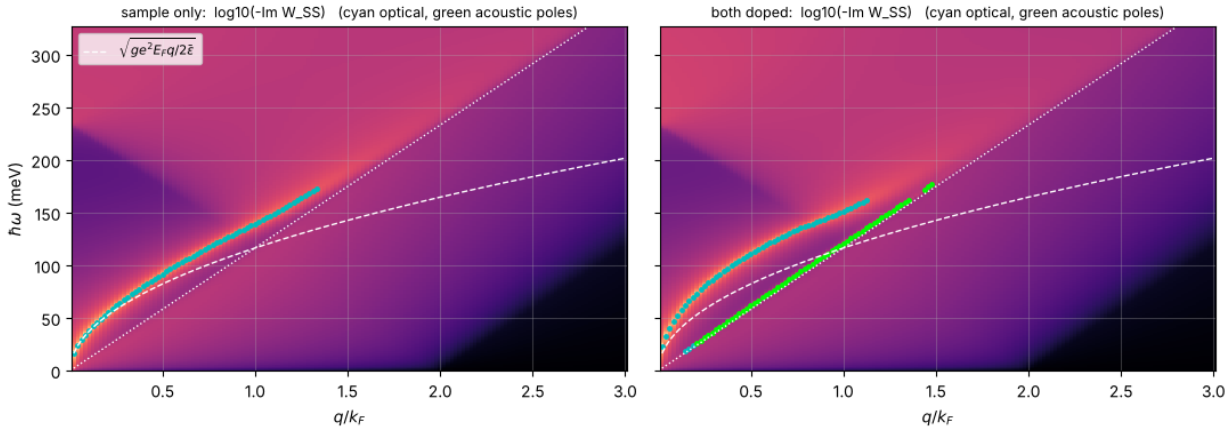


Figure 10: Loss $\log_{10}(-\operatorname{Im} W_{SS})$ of the sample sheet with the plasmon poles found as minima of $|\lambda_{\min}(\varepsilon)|$. Left: inert tip, one optical branch following $\sqrt{ge^2E_Fq/2\bar{\varepsilon}}$ (dashed) at small q. Right: both sheets doped, an additional acoustic branch (green) with $v_{\mathrm{ac}}/v_F = 1.048$, above the intraband continuum edge $\omega = v_Fq$ (dotted) as it must be. Both branches enter the continuum and stop being poles, and at 53% of the q points there is no pole at all.

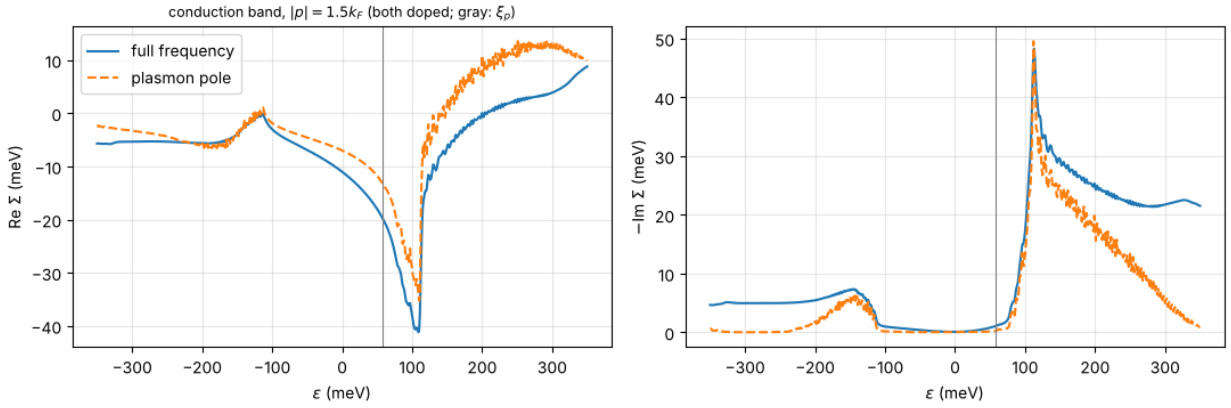


Figure 11: Σ of the conduction band at $|\boldsymbol{p}| = 1.5k_F$ (bare energy $\xi_{\boldsymbol{p}} = 58$ meV, gray). Full frequency (solid) vs plasmon-pole approximation (dashed). The PPA reproduces the plasmon-emission threshold at $\varepsilon \simeq 110$ meV but misses the continuum contribution to $-\operatorname{Im}\Sigma$ at high energy and the associated part of $\operatorname{Re}\Sigma$. Its ripples come from the discrete set of poles $(\boldsymbol{q}, \omega_p(\boldsymbol{q}))$ on the q grid.

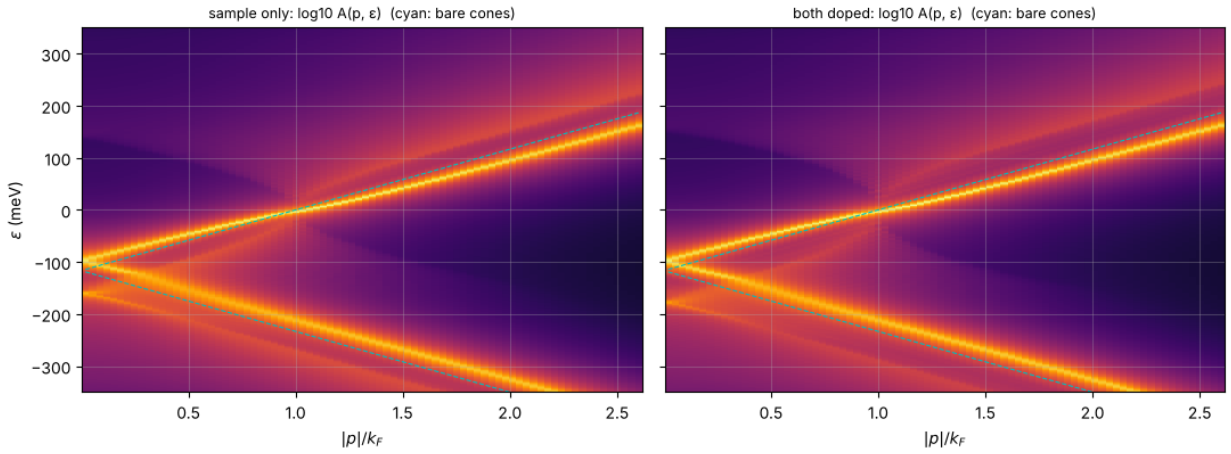


Figure 12: $\log_{10} A(|\boldsymbol{p}|, \varepsilon)$ of the sample, inert tip (left) and doped tip (right), with the bare cones in cyan. The quasiparticle bands are flattened by the correlation self-energy ($v^*/v_F = 0.92$, $Z \simeq 0.82$–0.83). The faint replica bands displaced away from E_F are the plasmaron satellites, the strongest one below the Dirac point at small $|\boldsymbol{p}|$. At $|\boldsymbol{p}| = 1.33k_F$ the window ± 350 meV holds 98.8% of the spectral weight.

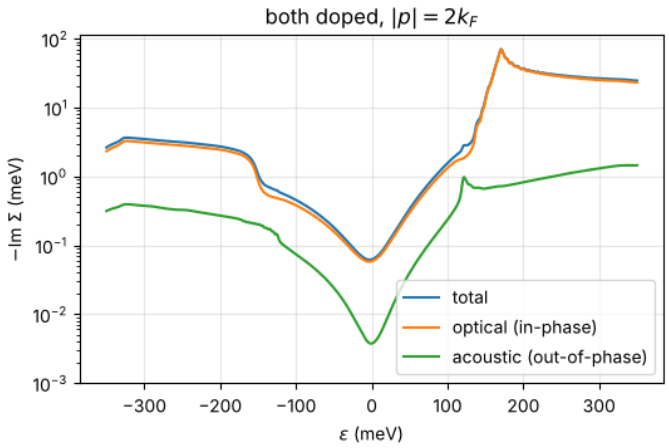


Figure 13: Layer-channel decomposition (eq. (21)) of $-\operatorname{Im}\Sigma$ at $|\boldsymbol{p}| = 2k_F$, both sheets doped. The split is exact (additivity 10^{-15}), and the acoustic (out-of-phase) channel carries $\simeq 4\%$ of $\|\Sigma\|$.

The acoustic plasmon of the junction contributes little to the sample self-energy at a 1 nm spacing because its weight in W_{SS} is small: the out-of-phase Coulomb channel $V_- \propto d$ is weak (section 2.4 and fig. 13).

8.3 Twisted bilayer graphene (notebook 03)

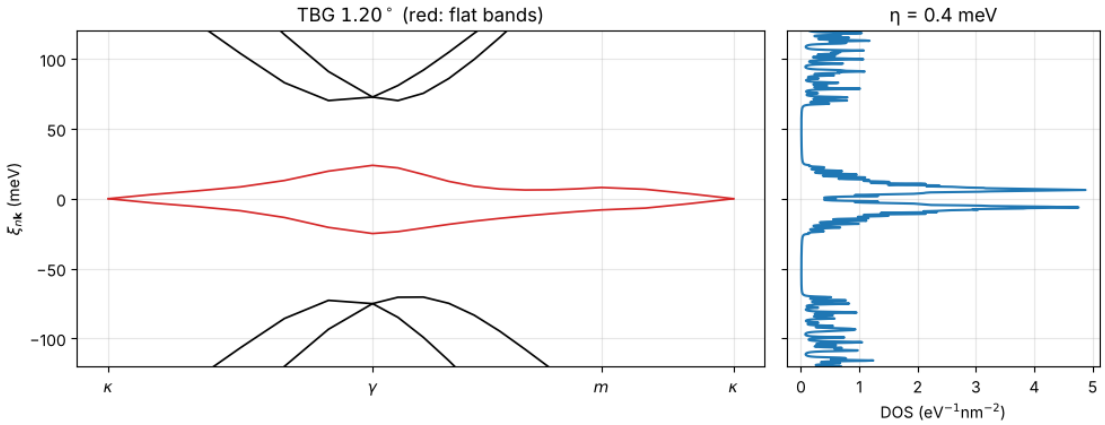


Figure 14: Continuum bands of TBG at 1.20° ($u_{AB} = 117$ meV, $u_{AA} = 0.7u_{AB}$) along κ–γ–m–κ of the 18×18 mesh, and density of states. Flat bands (red): widths 24.9 and 23.9 meV, gaps of 44 and 45.5 meV to the remote bands, chemical potential $\mu = +1.8$ meV at neutrality.

Figures 14 to 17 show the full chain for TBG. Two features deserve comment. (i) The satellites of the flat bands are dispersionless because both the flat-band plasmon and the intermediate flat-band states are nearly dispersionless. They are the structure the QTM is designed to detect (section 7.3). (ii) The band-off-diagonal self-energy changes the spectral function only where bands lie within $|\Sigma|$ of each other, but there the change is of order one. The band-diagonal approximation is therefore unreliable inside the flat-band manifold and at band crossings.

8.4 QTM on doped graphene (notebook 04)

The twist $\theta(k_F) = 0.60°$ brings the probed momentum to the Fermi surface, where the conduction-band ridge crosses $eV = 0$ (fig. 18). For $\theta \to 0$ both ridges meet at the Dirac point, at $eV \simeq -100$ meV instead of $-E_F = -117$ meV because of $\operatorname{Re}\Sigma$. The chiral matrix element changes the map by less than 0.7% with respect to a constant one, as expected from eq. (65). The satellites run

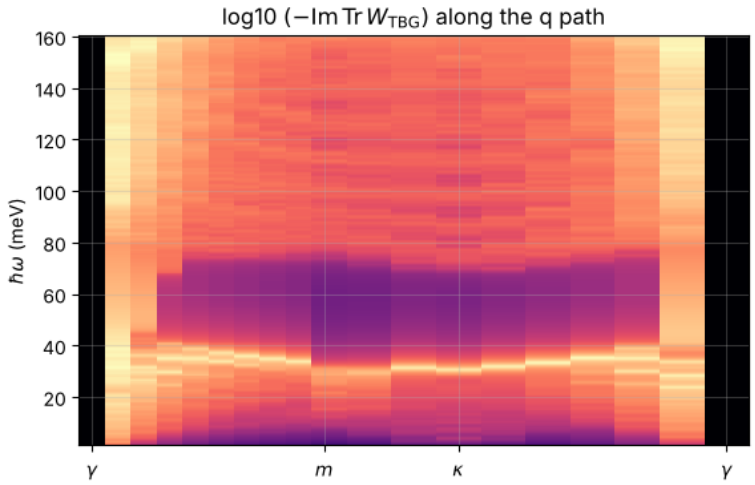


Figure 15: Loss function $\log_{10}(-\operatorname{Im}\operatorname{Tr} W_{\mathrm{TBG}})$ of the two TBG sheets in the SLG | 1 nm hBN | TBG stack (neutral tip), along γ–m–κ–γ ($\boldsymbol{q} = 0$ is excluded, black), with the valley-summed χ^0 of eq. (16). The nearly flat branch at 30–40 meV is the flat-band plasmon, which lies in the gap between the intra-flat-band and the flat-to-remote particle–hole continua [1].

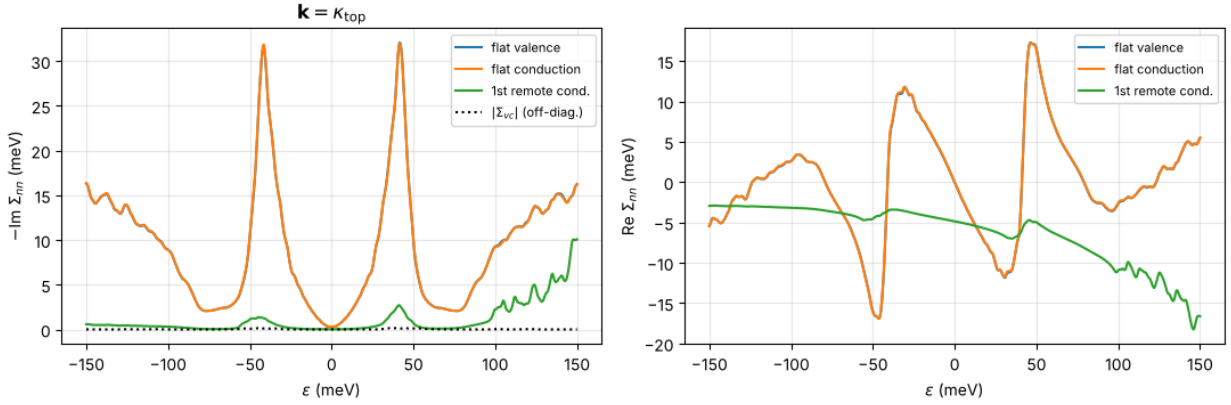


Figure 16: Band-matrix G_0W self-energy at $\boldsymbol{k} = \kappa_{\mathrm{top}}$ (diagonal elements, and dotted an off-diagonal element between the two flat bands, which are degenerate at κ). The peaks of $-\operatorname{Im}\Sigma$ of the flat bands at $\varepsilon \simeq \pm 42$ meV are plasmon emission, $\varepsilon \simeq \xi_{\mathrm{flat}} \pm \omega_{\mathrm{pl}}$, and $\operatorname{Re}\Sigma$ follows by Kramers–Kronig. Kernel and direct contraction agree to 7×10^{-5}, and Σ is passive.

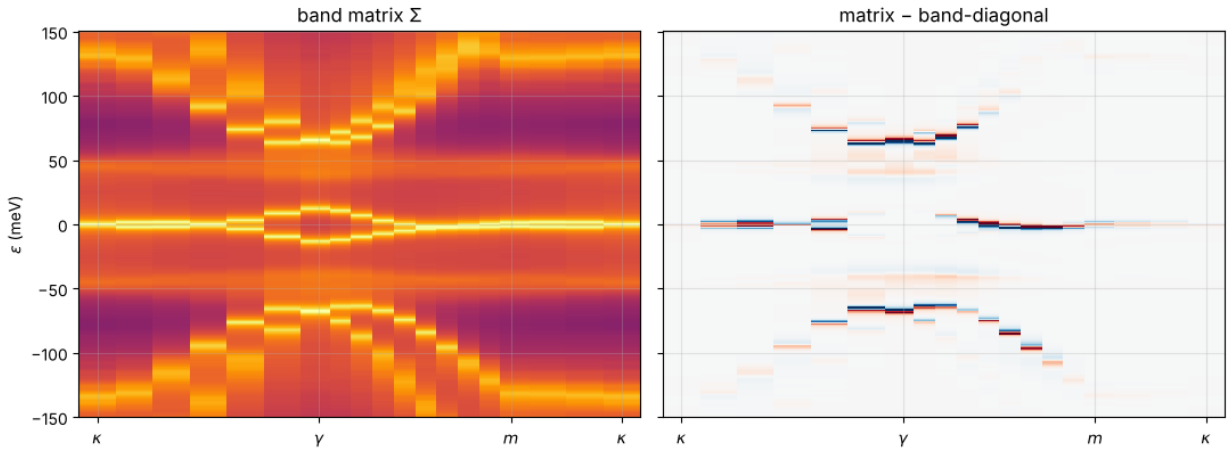


Figure 17: Left: $\log_{10} \operatorname{Tr} A(\boldsymbol{k}, \varepsilon)$ from the matrix Dyson equation ($\eta_{\mathrm{qp}} = 1$ meV) along κ–γ–m–κ. Bright lines: quasiparticle bands. The faint, nearly $\boldsymbol{k}$-independent lines at $\simeq \pm(40$–$50)$ meV are the plasmon satellites of the flat bands. Right: difference between the band-matrix and the band-diagonal $\operatorname{Tr} A$, which is concentrated where bands are close (flat bands, remote bands near γ).

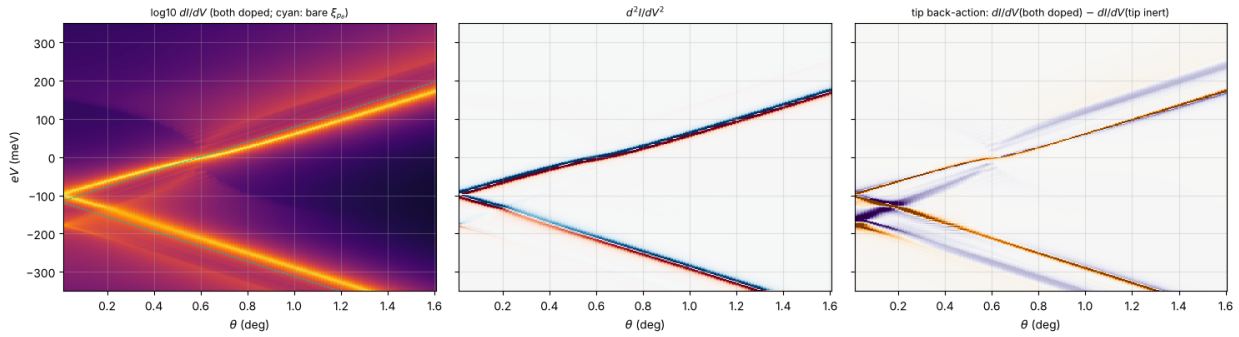


Figure 18: SLG tip | 1 nm hBN | SLG sample, both at 10^{12} cm^{-2}. Left: $\log_{10} dI/dV$ from eq. (65) (cyan: bare bands $\xi_{\pm}(\boldsymbol{p}_\theta)$). Middle: d^2I/dV^2 after 1.5 meV Gaussian smoothing. Right: tip back-action, the difference between a doped and an inert (non-screening) tip.

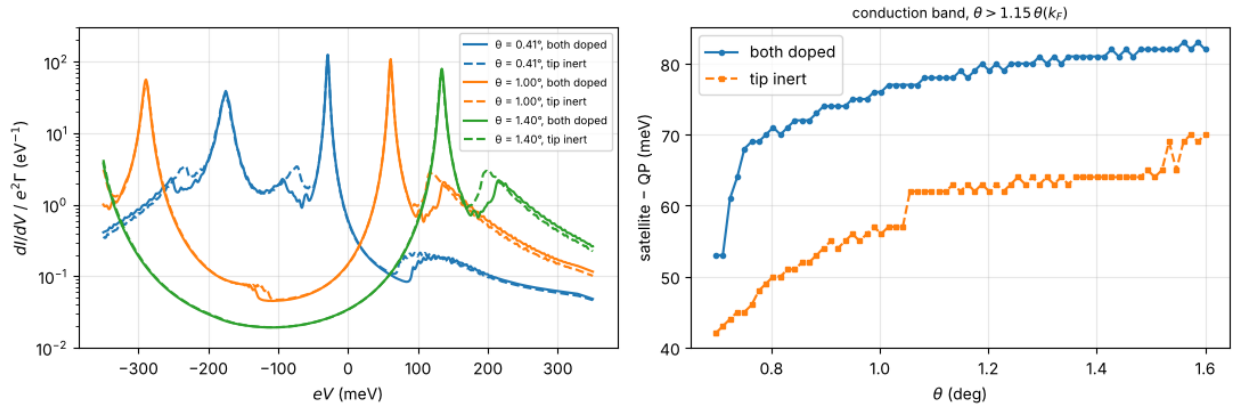


Figure 19: Left: line cuts of dI/dV at three twist angles, doped (solid) and inert (dashed) tip. Right: offset of the plasmaron satellite from the quasiparticle peak of the conduction band for $\theta > 1.15\,\theta(k_F)$. It saturates at $\simeq 80$ meV with a doped tip and $\simeq 63$ meV with an inert one.

parallel to the quasiparticle ridges on the side away from E_F. Their offset from the quasiparticle peak grows with θ and saturates (fig. 19). Doping the tip adds its Drude weight to the in-phase plasmon of the junction and raises the offset by $\simeq 17$ meV, the back-action that a QTM with a metallic tip necessarily has. The offset is not the plasmon energy at a single momentum, and its origin and size are explained in section 9.3.

8.5 QTM on twisted bilayer graphene (notebook 05)

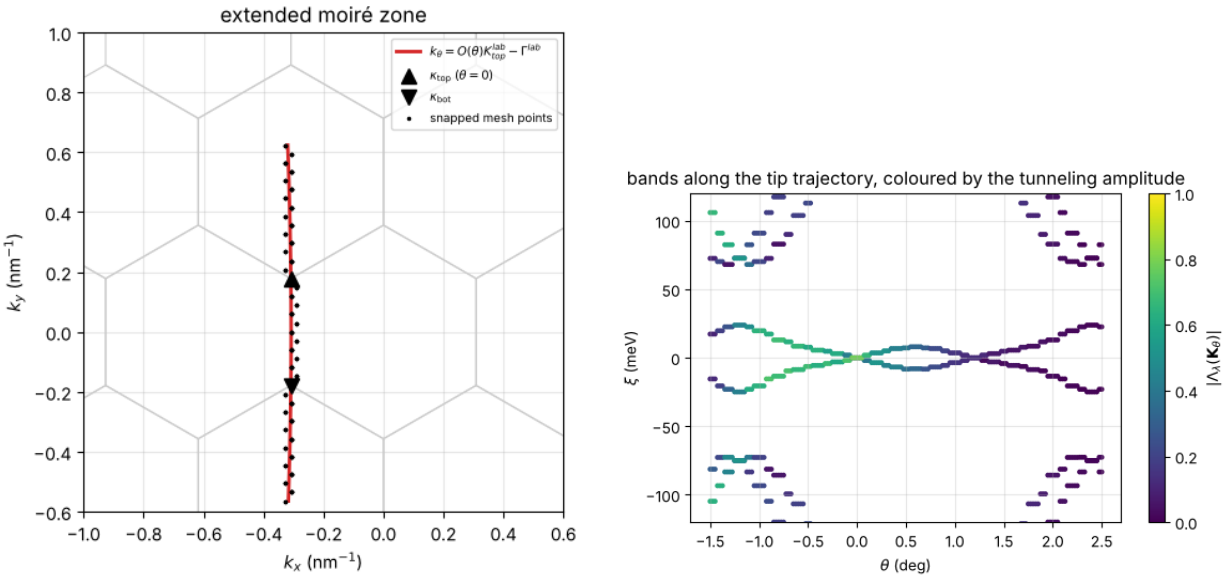


Figure 20: Left: tip trajectory $\boldsymbol{k}_\theta$, $\theta \in [-1.5°, 2.5°]$, in the extended moiré zone (red), with the mesh points it is snapped to. Right: the eight bands at the snapped momenta, coloured by $|\Lambda_\lambda(\boldsymbol{k}_\theta)|$ of eq. (68): near $\theta = 0$ the tip sees the flat bands through their top-layer Dirac-cone component, while away from κ_{top} the weight moves out of the computed band window.

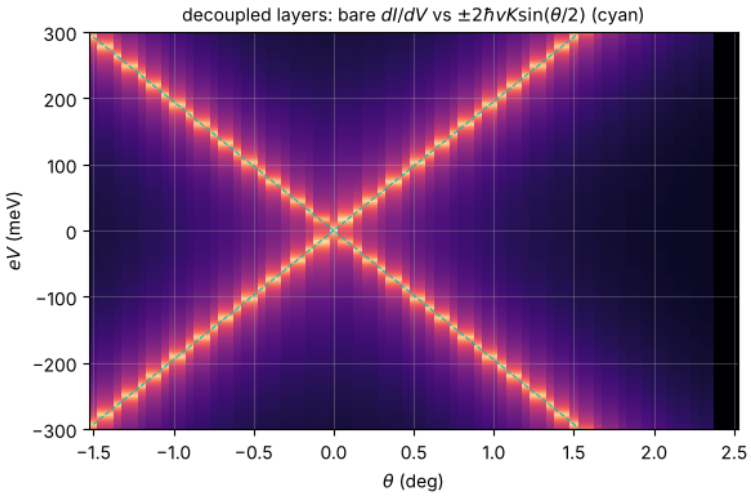


Figure 21: Validation with the interlayer hopping switched off (24×24 mesh, 16 bands): the bare conductance follows the top-layer Dirac cone $eV = \pm 2\hbar v K \sin(\theta/2)$ (cyan) for all θ, also outside the first moiré zone, because the amplitude is taken from the plane wave that unfolds the momentum (eq. (68)). The map is dark for $\theta \gtrsim 2.4°$, where $\boldsymbol{k}_\theta$ leaves the plane-wave cutoff.

The physical kinematics puts the Dirac points of the two layers at $\theta = 0$ and $\theta = \theta_{\text{TBG}}$, where the flat-band quasiparticle peak crosses $eV = 0$, and the γ point of the next zone at $\theta = -\theta_{\text{TBG}}$,

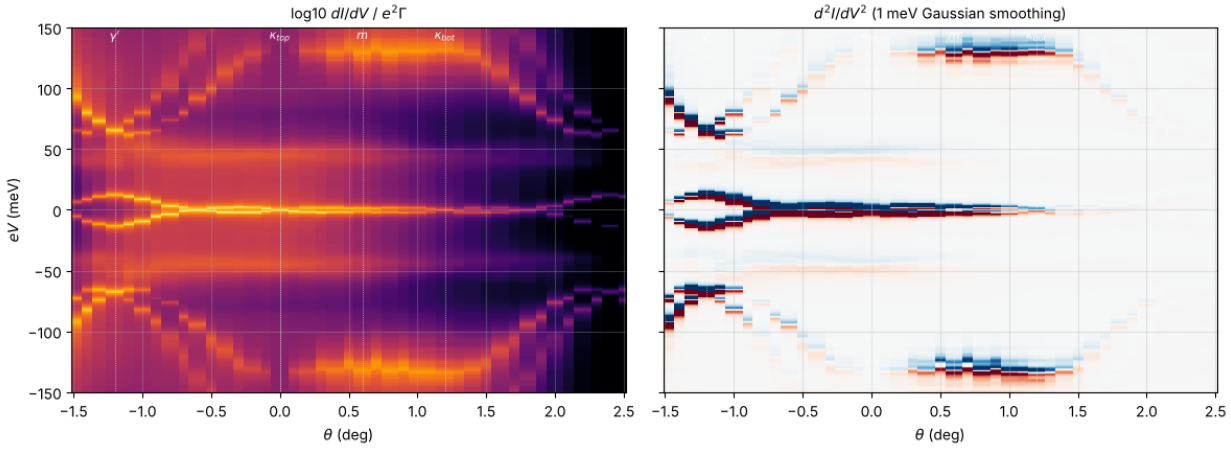


Figure 22: $dI/dV(V,\theta)$ (left, log scale) and d^2I/dV^2 (right, 1 meV smoothing) of the SLG tip | 1 nm hBN | TBG junction at neutrality, from the band-matrix G_0W spectral function. Dotted lines: θ at which the tip probes γ', $\kappa_{\rm top}$, m and $\kappa_{\rm bot}$. The map is piecewise constant in θ because the momentum is snapped to the 18×18 mesh (36 distinct points).

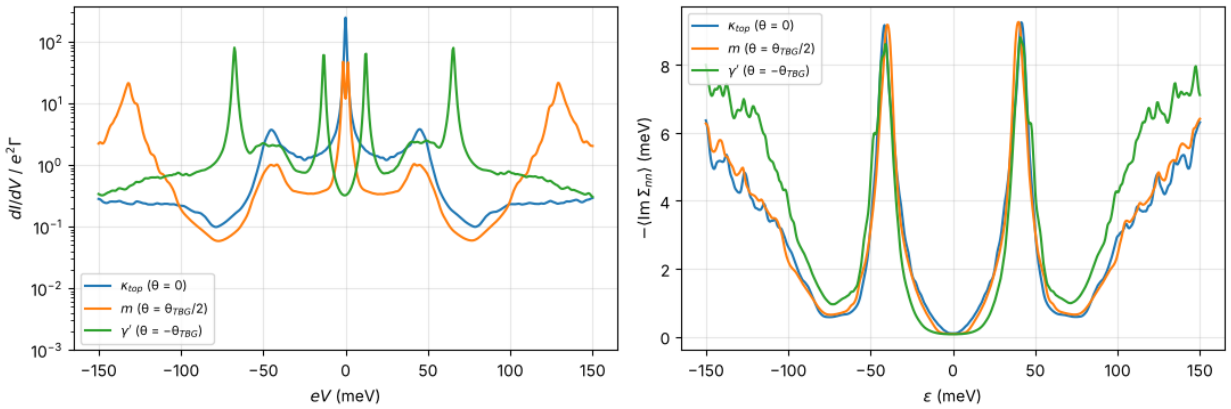


Figure 23: Left: dI/dV at $\kappa_{\rm top}$, m and γ'. Right: band-averaged $-\,\mathrm{Im}\,\Sigma_{nn}$ at the same momenta. The quasiparticle peaks move with θ; the satellites at $|eV| \simeq 45$ meV do not, because they come from the emission of the nearly dispersionless flat-band plasmon by flat-band electrons (the peaks of $-\,\mathrm{Im}\,\Sigma$ at ± 40 meV).

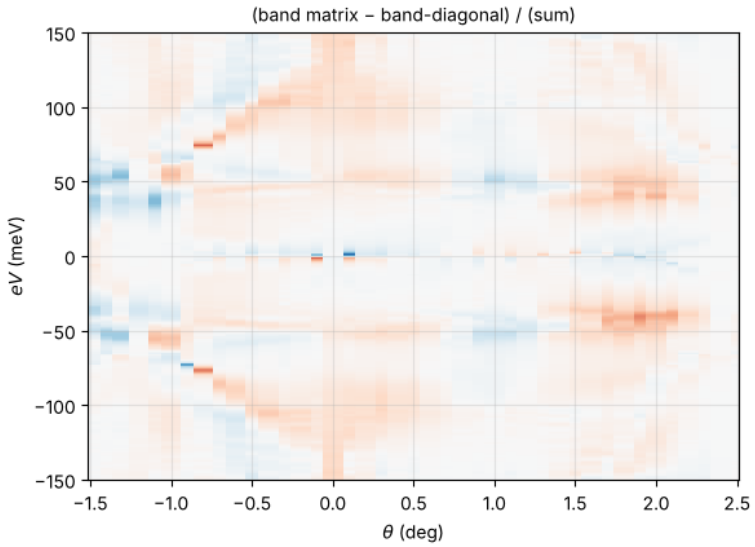


Figure 24: Relative difference between the conductance from the band-matrix spectral function and from the band-diagonal approximation, $(dI/dV_{\rm matrix} - dI/dV_{\rm diag})/(\mathrm{sum})$, along the trajectory.

where the flat bands are split by $\simeq 25$ meV and the remote bands are visible at ± 66 meV (figs. 22 and 23). The flat-band satellites appear as two horizontal stripes at $|eV| \simeq 45$ meV across the whole map. In d^2I/dV^2 they are weak compared with the quasiparticle edges. Beyond $\theta \simeq 1.8°$ the map darkens because the top-layer weight at $\boldsymbol{k}_\theta$ is carried by bands outside the eight computed ones. The band-off-diagonal self-energy changes the conductance by $\simeq 10\%$ in L^2 norm, locally by tens of percent where bands are close (fig. 24).

9 Interpretation of the results

This section explains what the features of the spectral functions and conductance maps of section 8 mean physically, and how the energy of the plasmon satellites can be estimated from the plasmon dispersion. The numbers are produced by notebook 06 and `g0w.satellites`.

9.1 What the spectral function contains

$A(\boldsymbol{k}, \varepsilon)$ is the probability density for adding an electron of momentum $\boldsymbol{k}$ at energy $\varepsilon > 0$ (or removing one at $\varepsilon < 0$), normalised as $\int \mathrm{d}\varepsilon\, A = 1$. In an interacting metal the added electron has two kinds of final state. The first is the **quasiparticle**, the electron dressed by a cloud of *virtual* density fluctuations. It gives the sharp peak at $E^*_{\boldsymbol{k}}$ (section 6), with weight $Z < 1$ ($Z \simeq 0.82$–0.83 in doped graphene), displaced from the bare band by $\mathrm{Re}\,\Sigma$. The second is an **electron plus a real plasmon**: the electron enters by creating a plasmon of momentum $-\boldsymbol{q}$ and ends at $\boldsymbol{k}+\boldsymbol{q}$ (fig. 6a). This costs $\varepsilon = \xi_{\boldsymbol{k}+\boldsymbol{q}} + \omega_{\mathrm{pl}}(\boldsymbol{q})$, so the weight $1 - Z$ is spread about one plasmon energy away from the band. Where these final states pile up, $-\mathrm{Im}\,\Sigma$ has a peak and A shows a faint replica of the band, the *satellite*. This is the electronic analogue of the phonon sidebands and shake-up satellites of photoemission, with the plasmon in place of the phonon. Two consequences are general. The satellite of an empty state lies above the quasiparticle peak and that of an occupied state below it: always on the side away from E_F. And the satellite is not a separate particle and has no fixed energy of its own: it is the edge of a continuum, whose position depends on the plasmon *dispersion* and on the band, as quantified next. At the level of G_0W only one satellite appears. Replicas at multiples of the plasmon energy require higher-order (vertex or cumulant) corrections [22].

9.2 One-plasmon picture of $-\mathrm{Im}\,\Sigma$

Inserting the plasmon part of the spectral density, $S(\boldsymbol{q}, \omega) \simeq R(\boldsymbol{q})\,\delta(\omega - \omega_{\mathrm{pl}}(\boldsymbol{q}))$, into the golden-rule form (47) of $-\mathrm{Im}\,\Sigma$ gives for an empty state ($\varepsilon > 0$, $T = 0$)

$$-\mathrm{Im}\,\Sigma_n(\boldsymbol{k}, \varepsilon) \simeq \pi \sum_{\boldsymbol{q}} w_{\boldsymbol{q}} \sum_m \Theta(\xi_{m,\boldsymbol{k}+\boldsymbol{q}})\, |\rho_{nm}(\boldsymbol{k}, \boldsymbol{q})|^2_R\, \delta(\varepsilon - E_{m\boldsymbol{k}}(\boldsymbol{q})), \qquad E_{m\boldsymbol{k}}(\boldsymbol{q}) = \xi_{m,\boldsymbol{k}+\boldsymbol{q}} + \omega_{\mathrm{pl}}(\boldsymbol{q}), \tag{69}$$

where $|\rho|^2_R = \sum_{ab} \rho^a_{nm} R_{ab} \rho^{b*}_{nm}$ (for graphene, the chirality factor times the residue). This is a density of final states on the $\boldsymbol{q}$ plane, weighted by the coupling. Like a density of states, it peaks where the final energy $E_{m\boldsymbol{k}}(\boldsymbol{q})$ is *stationary* in $\boldsymbol{q}$. The satellite energy is therefore a stationary value of $E_{m\boldsymbol{k}}(\boldsymbol{q})$, not ω_{pl} at a particular $\boldsymbol{q}$. Equation (69) is used here only for interpretation: all spectra of section 8 are computed with the full spectral density of W, which also contains the particle–hole continuum.

9.3 Dispersive plasmon: the emission edge in graphene

Estimate. At fixed $q = |\boldsymbol{q}|$, the direction of $\boldsymbol{q}$ sweeps the final band energy $\xi_{\boldsymbol{k}+\boldsymbol{q}}$ over $[\xi_{\boldsymbol{k}} - \hbar v_F q,\ \xi_{\boldsymbol{k}} + \hbar v_F q]$. The lowest value belongs to emission against the electron's motion (the electron loses momentum along its velocity), and the angular density diverges at both ends of the interval. The lower ends

$$E_-(q) = \xi_{\boldsymbol{k}} + \omega_{\mathrm{pl}}(q) - \hbar v_F q \tag{70}$$

form a curve that, for a gapless 2D plasmon, first rises and then falls. At its maximum a whole range of q starts contributing at the same energy (a caustic), the inverse-square-root singularities add up, and $-\operatorname{Im}\Sigma$ rises steeply. This is the emission edge:

$$\boxed{\varepsilon_{\rm sat} \simeq \xi_{\boldsymbol{k}} + \Delta, \qquad \Delta = \max_q \big[\omega_{\rm pl}(q) - \hbar v_F q\big], \qquad \left.\frac{d\omega_{\rm pl}}{dq}\right|_{q^*} = v_F} \tag{71}$$

(`satellites.dispersive_edge`). The maximum is reached where the plasmon group velocity equals the electron velocity, a Cherenkov-like matching condition. The final state must be empty, $\xi_{\boldsymbol{k}} - \hbar v_F q^* \geq 0$, i.e. $|\boldsymbol{k}| \geq k_F + q^*$, and for smaller $|\boldsymbol{k}|$ the edge is cut by Pauli blocking. For occupied states the mirror construction gives $\varepsilon_{\rm sat} \simeq \xi_{\boldsymbol{k}} - \Delta$ for $|\boldsymbol{k}| \leq k_F - q^*$. Below the edge only small q (small weight $w_q R$) and large q (Landau-damped plasmon) contribute, which is why $-\operatorname{Im}\Sigma$ is small there.

Closed form. For the long-wavelength plasmon $(\hbar\omega_{\rm pl})^2 = \frac{g}{2}\frac{e^2}{\bar\varepsilon} E_F q \equiv a^2 q$, maximising $a\sqrt{q} - \hbar v_F q$ gives

$$q^* = \frac{a^2}{4(\hbar v_F)^2} = \frac{g}{8}\,\alpha_g\, k_F, \qquad \Delta = \frac{a^2}{4\hbar v_F} = \frac{g}{8}\,\alpha_g\, E_F \overset{g=4}{=} \frac{\alpha_g E_F}{2}, \qquad \alpha_g = \frac{e^2}{\bar\varepsilon \hbar v_F} \tag{72}$$

(`satellites.sqrt_plasmon_edge`): the satellite offset is a fraction of E_F set by the graphene fine-structure constant. For our junction ($\alpha_g = 0.50$, $E_F = 116.6$ meV) this gives $\Delta = 29$ meV and $q^* = 0.25 k_F$. The RPA plasmon of the junction lies above the $\sqrt{q}$ law at $q \sim 0.4 k_F$, and the full dispersion gives larger values (table 1).

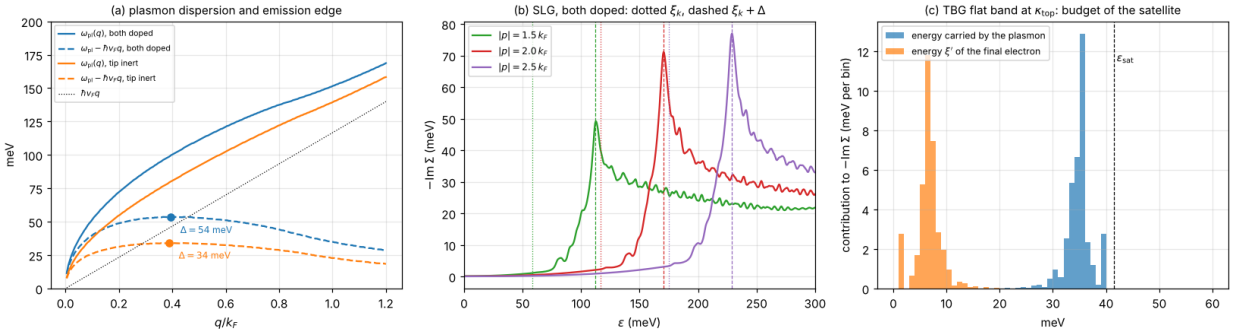


Figure 25: (a) RPA plasmon dispersion of the sample sheet (solid) and $\omega_{\rm pl}(q) - \hbar v_F q$ (dashed) for the doped and the inert tip, with dots marking the maximum Δ of eq. (71). (b) Full-frequency $-\operatorname{Im}\Sigma$ of the conduction band at three momenta (both sheets doped): the edge sits at $\xi_{\boldsymbol{k}} + \Delta$ (dashed), one constant offset above the bare band (dotted). (c) TBG, flat conduction band at $\kappa_{\rm top}$: decomposition of $-\operatorname{Im}\Sigma$ at its peak $\varepsilon_{\rm sat} = 41.5$ meV into the energy carried by the plasmon and the energy of the final electron.

Table 1: Emission edge of the graphene sample: eq. (71) against the full-frequency calculation. All energies in meV, E^* is the quasiparticle peak of A.

tip	Δ	q^*/k_F	$\lvert\boldsymbol{k}\rvert/k_F$	$\xi_{\boldsymbol{k}}$	$\xi_{\boldsymbol{k}} + \Delta$	peak of $-\operatorname{Im}\Sigma$	satellite of A	sat. $- E^*$
doped	53.7	0.40	1.5	58.3	112.0	112.5	115.0	73.0
			2.0	116.6	170.3	170.5	174.5	79.0
			2.5	174.9	228.6	229.0	233.0	82.0
inert	34.1	0.39	1.5	58.3	92.4	92.5	95.5	53.5
			2.0	116.6	150.7	151.0	157.5	62.5
			2.5	174.9	209.0	209.0	215.0	64.5

Check. Equation (71) reproduces the peak of the full-frequency $-\operatorname{Im}\Sigma$ to within 0.5 meV at every momentum (table 1 and fig. 25b), and the plasmon slope at q^* equals v_F to 2%. The satellite peak of A lies 2–7 meV higher, pushed by the Dyson denominator $(\varepsilon - \xi_{\boldsymbol{k}} - \operatorname{Re}\Sigma)^2$ of eq. (56).

Energy budget. Neither $\omega_{\mathrm{pl}}(q^*)$ ($\simeq 100$ meV) nor Δ ($\simeq 54$ meV) is the energy of the emitted plasmon. Decomposing $-\operatorname{Im}\Sigma$ at the edge into the energy $\omega = \varepsilon - \xi'$ carried away by the plasmon and the energy ξ' of the final electron (`satellites.slg_energy_loss`, $\eta \to 0$): at $|\boldsymbol{k}| = 1.5k_F$ the plasmon carries on average 82 meV (inter-quartile range 67–100 meV) and the electron ends at $\xi' \simeq 30$ meV, with momentum transfers around $0.3k_F$. At $2k_F$ the numbers are 96 meV (77–115 meV) and $\xi' \simeq 75$ meV. The energy of the added electron is shared between the plasmon and the drop of the electron's own band energy.

Offset seen in a QTM line cut. A line cut measures the satellite relative to the quasiparticle peak,

$$\varepsilon_{\mathrm{sat}} - E^*_{\boldsymbol{k}} \simeq \Delta + (\xi_{\boldsymbol{k}} - E^*_{\boldsymbol{k}}) + \delta_D, \tag{73}$$

where $\xi_{\boldsymbol{k}} - E^*_{\boldsymbol{k}} = -\operatorname{Re}\Sigma$ at the quasiparticle (16, 21 and 24 meV at $|\boldsymbol{k}| = 1.5$, 2 and $2.5k_F$) and $\delta_D \simeq 2$–7 meV is the Dyson shift. This accounts for the offsets of fig. 19:

- **Plateau.** Δ does not depend on $\boldsymbol{k}$, so the offset grows only through the slowly increasing $-\operatorname{Re}\Sigma$ and is nearly constant at large θ: $\simeq 80$ meV with a doped tip, $\simeq 63$ meV with an inert one.
- **Onset.** The edge needs $|\boldsymbol{k}| \geq k_F + q^* \simeq 1.4k_F$, i.e. $\theta \gtrsim 1.4\,\theta(k_F) = 0.83°$ (eq. (64)). Below that the edge is Pauli-blocked and the offset is smaller, as in the rising part of the curve.
- **Tip back-action.** Doping the tip raises Δ from 34.1 to 53.7 meV. Together with the slightly larger Dyson shift of the inert case this gives the 17 meV increase of the QTM offset. The tip's carriers oscillate in phase with the sample's in the optical plasmon of the junction, adding Drude weight and stiffening its dispersion (in the limit $qd \to 0$ the in-phase mode of two identical sheets has twice the Drude weight, $a^2 \to 2a^2$, which would double Δ, while at q^* the computed ratio is 1.57). The satellite energy therefore depends on the electrostatic environment of the sample, including the probe.

Because Σ contains only the correlation part (section 5.12), the quasiparticle shift in eq. (73) is correlation-only. The Fock term would move E^* but not $\varepsilon_{\mathrm{sat}}$.

9.4 Flat plasmon and flat bands: twisted bilayer graphene

In TBG at 1.2° both the plasmon [23, 1] and the final states are nearly dispersionless: the peak of the sample spectral density is at 34.2 meV over the whole moiré mesh (inter-quartile range 32.7–35.2 meV), and the flat bands are $\simeq 25$ meV wide. The $\boldsymbol{q}$ integral then hardly matters, and the stationary value of $E_{m\boldsymbol{k}}(\boldsymbol{q})$ is reached where the density of final flat-band states peaks, at the van Hove energy ξ_{VH}:

$$\boxed{\varepsilon_{\mathrm{sat}} \simeq \pm(\xi_{\mathrm{VH}} + \bar{\omega}_{\mathrm{pl}})} \tag{74}$$

With $\xi_{\mathrm{VH}} = 6.2$ meV (peak of the conduction flat-band DOS) this gives 40.5 meV, against the computed peak of $-\operatorname{Im}\Sigma$ of the flat band at κ_{top}, 41.5 meV. The decomposition (`satellites.tbg_energy_loss`, fig. 25c) confirms the picture: the plasmon carries 34.8 meV on average (inter-quartile range 33.8–35.6 meV) and all final states lie in the flat band, at $\xi' \simeq 6.7$ meV. The satellite of A and of dI/dV appears at $|eV| \simeq 45$ meV, again a few meV beyond the peak of $-\operatorname{Im}\Sigma$ because of the Dyson denominator. The hole satellite is the mirror image, by the approximate particle–hole symmetry of the continuum bands.

Since neither $\bar{\omega}_{\mathrm{pl}}$ nor ξ_{VH} depends on $\boldsymbol{k}$, the satellite energy is the same at every momentum probed by the tip. This is why, in figs. 22 and 23, the satellites form horizontal stripes that do not move with the twist angle, while the quasiparticle peaks follow the bands along the trajectory (zero

bias at $\kappa_{\rm top}$ and $\kappa_{\rm bot}$, ± 12 meV at γ'). The angle independence is the fingerprint that distinguishes a flat-plasmon satellite from a band.

9.5 What a QTM measurement of the satellites determines

Since $dI/dV \propto \langle\chi_\theta|A(\boldsymbol{K}_\theta, eV)|\chi_\theta\rangle$ (eq. (62)), the quasiparticle ridges of a map give the renormalised bands and the satellites give the one-plasmon emission edges. In graphene the satellite offset is not the plasmon energy but $\Delta = \max_q[\omega_{\rm pl} - \hbar v_F q]$, a property of the plasmon dispersion near the momentum where its slope equals v_F, plus the quasiparticle shift of eq. (73). Extracting plasmon information from it requires either the measured quasiparticle ridge and band velocity, or the calculation. In flat-band TBG the offset is close to $\bar{\omega}_{\rm pl} + \xi_{\rm VH} - E^*_{\boldsymbol{k}}$, a nearly direct, angle-independent measure of the flat-band plasmon energy. In both cases the edge is sharpened in d^2I/dV^2, and its position depends on the screening environment, the tip included.

10 Numerics, validation and limitations

10.1 Grids and broadenings

	SLG | hBN | SLG (notebooks 02, 04)	SLG | hBN | TBG (notebooks 03, 05)
momenta $\boldsymbol{q}$	radial, $n_q = 144$ (02), 96 (04) on $[0.02, 3]k_F$; $n_\theta = 64$ angles	18×18 moiré mesh (324 points), $\boldsymbol{q} = 0$ removed
bands	two Dirac bands, analytic	8 continuum bands, $G_{\rm max} = 3$ (37 plane-wave shells)
ω grid for S	480 points on $[10^{-4}, 2.8E_F]$	320 points on $[2\eta, 160$ meV$]$
η (χ^0, V, W)	1 meV	0.71 meV $= 2\hbar v_{\rm BM}\lvert\boldsymbol{G}_M\rvert/N_s$
η_Σ (resolvents)	1 meV	$= \eta$
$\eta_{\rm qp}$ (Dyson)	2 meV	0.5 meV
ε grid ($= eV$)	701 points on ± 350 meV	601 points on ± 150 meV
kernel step du	$\eta_\Sigma/8$	$\eta_\Sigma/8$

Three broadenings, three roles. η turns the undamped collective modes into Lorentzians that the ω grid can integrate. It must exceed the grid step and, for TBG, the mesh floor $\hbar v_{\rm BM}|\boldsymbol{G}_M|/N_s$ (the energy a band of velocity $v_{\rm BM}$ sweeps across one mesh spacing) below which χ^0 resolves individual $\boldsymbol{k}$ points (`TBGParams.eta_mesh`). η_Σ regularises the resolvents of eq. (43). It sets the energy resolution of Σ and the floor of $-\operatorname{Im}\Sigma$ at $\varepsilon = 0$. $\eta_{\rm qp}$ is an extrinsic width added in the Dyson equation. It keeps A finite where $\operatorname{Im}\Sigma \to 0$ and plays the role of the experimental resolution. Physical widths are those of $-\operatorname{Im}\Sigma$ that exceed all three.

Away from the magic angle. The TBG heuristic $\eta = 2\hbar v_{\rm BM}|\boldsymbol{G}_M|/N_s$ scales with the renormalised velocity, and at 5° it gives ~ 120 meV and must be replaced by `eta_override`.

10.2 Cost

On one CPU core the TBG pipeline of notebook 03 takes 10 s for the band structure and 35 s for the form factors $\rho^\ell_{nm}(\boldsymbol{k}, \boldsymbol{q})$, which occupy 0.21 GB ($2 \times 8 \times 8 \times 324 \times 324$ complex numbers). χ^0 and W on 324×320 $(\boldsymbol{q}, \omega)$ points take 39 s and the resolvent kernel 4 s. The band-matrix Σ ($8 \times 8 \times 601$ values) then costs $\simeq 1$ s per $\boldsymbol{k}$ with the kernel, against $\simeq 12$ s for the direct contraction. The direct route scales as $n_q\, n_\omega\, n_b^2\, n_m\, n_\varepsilon$, the kernel route as $n_q\, n_b^2\, n_m\, n_\varepsilon$ after a one-off tabulation of cost $n_q\, n_\omega\, n_u$. For SLG the kernel route costs $\simeq$ 1–2 s per $(\boldsymbol{k}, \text{band})$ with $n_q = 96$–144 and $n_\vartheta = 64$, against $\simeq 80$ s for the direct integration. Memory is dominated by `densityL`, which grows as N_s^4: $N_s = 24$ needs 0.7 GB and $N_s = 30$ 1.7 GB.

10.3 Validation

The test suite (`python tests/run_tests.py`, 42 tests, ∼1.5 min) checks:

check	result
V of 1 and 2 sheets, gated and ungated, against closed forms (section 2)	$< 10^{-6}$ relative
capped-slab limits; gate suppresses the in-phase channel more than the out-of-phase one	exact limits
$(\ell, \boldsymbol{G})$ flattening identical in ε and V	exact
closed-form graphene χ^0 vs brute-force Lindhard sum (1000×1000 polar grid)	$< 2 \times 10^{-3} D(E_F)$
$\chi^0(q \to 0, 0) = -D(E_F)$; $\operatorname{Im}\chi^0 \leq 0$ for $\omega > 0$	yes
fast TBG χ^0 vs `pybandstructure polarization`	3×10^{-15}
TBG χ^0 and W obey the Onsager relation $X(\boldsymbol{q}) = X(-\boldsymbol{q})^T$	5×10^{-16}
$\sum_\ell \rho^\ell_{nm}(\boldsymbol{k}, 0) = \delta_{nm}$	10^{-14}
band-matrix Σ: diagonal equals the independent band-diagonal routine	4×10^{-16}
kernel route vs direct contraction (SLG, TBG; notebooks 02, 03)	1.3×10^{-4}, 7×10^{-5}
Σ passive ($(\Sigma - \Sigma^\dagger)/2\mathrm{i} \preceq 0$), gauge covariant	yes
$\int \operatorname{Tr} A = n_b$ for $\Sigma = 0$; Dyson root and Z for linear Σ	10^{-3}; exact
channel split of W and Σ is additive	10^{-15}
TBG flat bands at 1.20°; model-frame Dirac points; eigenvector order	yes
tip trajectory through κ_{top}, κ_{bot}, γ'; snapping $\leq 1/\sqrt{3}$ spacing	yes
decoupled layers: bare dI/dV follows $\pm 2\hbar v K \sin(\theta/2)$	within 1.5 meV
band-matrix dI/dV gauge invariant and ≥ 0	yes
emission edge of a $\sqrt{q}$ plasmon equals eq. (72); slope v_F at q^*	10^{-6}; 1%
energy-loss decomposition sums to the full-frequency $-\operatorname{Im}\Sigma$	$< 5\%$

10.4 Limitations

- **Head only.** Local fields are omitted (assumption 4). For TBG at 1.2° the moiré umklapp $\boldsymbol{G} \neq 0$ components of χ^0 are not negligible at $|\boldsymbol{q}| \sim |\boldsymbol{G}_M|$, and momentum transfers outside the first zone are dropped from Σ. `LayerHelpers` supports $n_G > 1$ for χ^0 and ε, but the self-energy routines contract only the head.
- **No exchange.** Σ contains correlation only (section 5.12). For TBG at neutrality this matches the scheme of Ref. [1], while for doped graphene it misses the velocity enhancement.
- **Truncations.** For SLG, $q \leq 3k_F$: the large-q interband tail of W is cut off, which mainly affects $\operatorname{Re}\Sigma$ (logarithmically) and hence v^*, while the satellites, which come from $q \lesssim k_F$ (fig. 7b), are converged. For TBG, $\omega \leq 160$ meV and 8 bands: transitions to and between remote bands above this window are missing in S, and their effect on the flat bands is a nearly static real shift.
- **Mesh.** The TBG $\boldsymbol{q}$ sum is a 18×18 (or 12×12 for quick runs) mesh. Σ inherits the mesh ripples of χ^0 unless $\eta \gtrsim \eta_{\mathrm{mesh}}$, and the QTM trajectory is snapped to the mesh, so the (θ, V) map is piecewise constant in θ on steps of $|\boldsymbol{G}_M|/(N_s K)$ (0.12° for $N_s = 18$).
- **Band window.** Only the 8 computed bands enter A^S and the sum over intermediate states m. At large $|\theta|$ the tip probes momenta far from κ_{top} where the top-layer weight is carried by bands outside the window, and the map darkens.
- **Static dielectrics.** hBN is used with its static anisotropic constants unless a dispersive model from `g0w.dielectrics` is chosen. The phonon polaritons of hBN then enter $V(\boldsymbol{q}, \omega)$, but they are untested in the self-energy routines.
- **QTM.** Equation (62) assumes perfect momentum resolution and a constant tip DOS (section 7). Temperature and lock-in broadening are mimicked by Gaussian smoothing.

A Polarisability of doped graphene in closed form

At $T = 0$ and electron density n ($k_F = \sqrt{4\pi n/g}$), with $D_\pm = (\omega/\hbar v_F \pm 2k_F)/q$ and $s = \sqrt{\omega^2 - (\hbar v_F q)^2}$ (principal root), the retarded bubble of eq. (15) for one sheet is [8]

$$\chi^0(q,\omega) = -\frac{g}{16\pi}\Big[\frac{8k_F}{\hbar v_F} + \frac{q^2}{s}\big(\mathcal{G}(D_-) + \mathrm{i}\pi - \mathcal{G}(D_+)\big)\Big],$$
$$\mathcal{G}(z) = z\sqrt{z-1}\sqrt{z+1} - \log\big(z + \sqrt{z-1}\sqrt{z+1}\big) \tag{75}$$

(`dirac_chi.chi_dirac_phys`). The split root makes $\mathcal{G}$ analytic on $\mathbb{C} \setminus [-1,1]$, which is the cut structure of the retarded response, so the same expression is valid at real $\omega + \mathrm{i}0$ and at complex $\omega + \mathrm{i}\eta$. No branch switch is needed: $\mathcal{G}(-z) = \mathcal{G}(z) + \mathrm{i}\pi$ in the upper half plane, so the two forms $\mathcal{G}(-D_-)$ and $\mathcal{G}(D_-) + \mathrm{i}\pi$ that appear in the literature are the same analytic function there (checked to 3×10^{-14}). Limits: $\chi^0(q \to 0, 0) = -D(E_F)$ with $D(E_F) = gk_F/(2\pi\hbar v_F)$ the density of states at the Fermi level. For $\omega \ll \hbar v_F q$ and $q \ll k_F$ one recovers the static Thomas–Fermi form, and for $q \ll k_F$ and $\omega \gg \hbar v_F q$ the Drude form $\chi^0 \simeq D(E_F)\,(\hbar v_F q)^2/2\omega^2$, which with $V = 2\pi e^2/\bar{\varepsilon} q$ gives the optical plasmon $\omega_{\mathrm{pl}}^2 = \frac{g}{2}\frac{e^2}{\bar{\varepsilon}} E_F q$ (`test_optical_plasmon_long_wavelength_law`). The intrinsic (undoped) part $\chi^0 = -\frac{g}{16} q^2/\sqrt{(\hbar v_F q)^2 - \omega^2}$ is used for a neutral tip.

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